

PAIRS TRADING AND ECONOMETRIC TIME SERIES METHODS
APPLIED TO ENERGY EQUITIES

by

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DEDICATION

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LIST OF ABBREVIATIONS

Abbreviation	Description
ADF	Augmented Dickey and Fuller
COP	ConocoPhillips
CVX	Chevron Corporation
E&P	Exploration and Production
ETF	Exchange Traded Fund
IRF	Impulse Response Function
KPSS	Kwiatkowski, Phillips, Schmidt and Shin
S&P 500	Standard and Poor's 500 Index
VAR	Vector Autoregression
VaR	Value-at-Risk
XLE	Energy Select Sector SPDR Fund
XOM	Exxon Mobil Corporation

ABSTRACT

Statistical Arbitrage is a method used in finance to identify two cointegrated securities and profits from temporary mispricing between them relative to their historical spread. Replicating a study by Salas Najera [1], which applies cointegration based pairs trading to oil equities, I try to identify a cointegrated pair using data from 1999 to July 2026. While my analysis fails to find a strong cointegrated pair over this extended period, I retained the pairs used in the original study, ConocoPhillips (COP) and Exxon Mobil Corporation (XOM), which surprisingly delivered positive risk adjusted returns across market regimes. Its edge is primarily attributed to the variance reduction of a hedged position rather than a mean reversion pair trade. I also find that this edge decays in the most recent period of the sample.

1. INTRODUCTION

Pairs trading, also described as statistical arbitrage, was further developed at Morgan Stanley during the 1980s under the direction of Nunzio Tartaglia’s quantitative research group [2]. Its premise is that certain securities are bound together by economic fundamentals in a way that restricts how far their prices can drift apart each other. When the gap between two such securities diverges beyond its historical range, the strategy sells short the overperforming (expensive) leg and buys the relatively underperforming (cheap) leg. Profit is then realised as the spread reverts to its mean, provided the underlying economic relationship between the two assets remains constant.

1.1. Objectives and Rationale

This paper replicates the study by Salas Najera [1] which applies the cointegration method to two integrated oil majors (XOM and Chevron Corporation (CVX)), two Exchange Traded Funds (Energy Select Sector SPDR Fund (XLE) and Standard and Poor’s 500 Index (S&P 500)) and one exploration and production company (COP). This replication makes extensions to the original study: the programming language implementation, the sample period and additional statistical methods. The original study was conducted in Python over a period from 31 December 1999 to 31 December 2018. This replication is implemented in R and extends the sample to 24 July 2026.

This paper serves two objectives. The first is pedagogical. Re-implementing a strategy from its written description forces engagement with subtle methodological decisions that as readers we often skim or accept uncritically. The second is present day relevance. Since the original dataset ends in December 2018, the period from January 2019 onward represents a true out of sample period that the original author could not have observed or used for selecting a pair or parameter tuning. Most importantly, the additional sample period spans extraordinary market swings including the collapse of crude oil prices in early 2020 triggered

by COVID-19, Russia-Ukraine war in 2022 and geopolitical escalations between the United States and Iran in 2026 [3] that fundamentally altered the market dynamics of many securities in the global market.

2. DATA

2.1. Sample

The sample consists of three common stocks and two Exchange Traded Fund (ETF)s observed at a daily frequency from 31 December 1999 to 24 July 2026. This gives 6,680 price observations and 6,679 return observations per series with no missing values.

The three equities are COP, CVX and XOM. CVX and XOM are integrated energy firms operating across exploration, refining and distribution. In contrast, COP is a pure play Exploration and Production (E&P) company following its 2012 downstream spin-off [4]. This structural distinction in business models matters when evaluating cointegration across equity pairs. A pure exploration and production firm is levered almost entirely to raw crude oil prices. In contrast, an integrated firm maintains downstream refining capacity which benefits from expanding crack spreads (the margin between raw crude input costs and refined product prices) providing an almost natural hedge when crude prices drastically fall.

Additionally, two ETFs are included as a control rather than pair trading candidates. The xle tracks the energy component of the S&P 500 and serves as the sector benchmark. Meanwhile, the (SPY) tracks the broader market index and represents the aggregate market (systematic) risk across the broader economy.

2.2. Data Source and Adjustment

Prices were retrieved from Yahoo Finance through the `quantmod` package in R, which follows the standard CRSP-style adjustment methodology. Throughout this study, adjusted closing prices are used in place of raw closing prices. This was done to capture total returns by accounting for stock splits and cash dividends.

Using adjusted prices is a very deliberate choice especially because of the use of energy equities. Over a 26-year sample period (1999–2026), integrated oil companies historically maintained high dividend yields (typically 3%–5% annually), creating a substantial cumu-

lative divergence between price appreciation and total return [5]. Secondly, the adjustment is important for cointegration testing. Unadjusted prices introduce discrete downward price jumps on ex dividend dates. Since candidate pair equities pay dividends on different corporate calendars, these unadjusted drops do not cancel out symmetrically. If left uncorrected, they would generate artificial spread dislocation and false trading signals.

While the original study does not explicitly specify whether raw or adjusted prices were used, my study employs adjusted closing prices and notes the importance of this implementation choice.

2.3. Return Construction

Returns are computed as continuously compounded log returns,

$$r_t = \ln P_t - \ln P_{t-1} \tag{2.1}$$

where P_t denotes the adjusted close on day t . Log returns are preferred to simple returns for two reasons. They are additive across time enabling cumulative returns to be computed as the simple sum of daily returns. Also, log returns exhibit greater symmetry around zero because they are unbounded on both tails $(-\infty, +\infty)$, while simple returns are truncated below at -100% . Calculating first differences uses one initial observation, leaving 6,679 return observations per series.

2.4. Sample Period and Macroeconomic Context

The extension of the sample beyond the original 2018 endpoint introduces episodes that warrant explicit acknowledgement, since they bear weight on the interpretation of results.

In March and April 2020 the collapse in global oil demand produced the first negative settlement in the history of the West Texas Intermediate futures contract [6] and energy equities fell by more than half within weeks before recovering strongly through 2022. Any

parameter estimated on data ending in 2013, as the original study's trading thresholds are, is unlikely to describe this regime well.

Separately, the identity of the underlying firms changed. Exxon Mobil completed its acquisition of Pioneer Natural Resources in 2024 [7] and ConocoPhillips completed its acquisition of Marathon Oil in the same year [8], in both cases substantially enlarging their Permian Basin positions.

Finally, the weight of the energy sector within the S&P 500 fell to roughly 2% by 2020 before partially recovering, so SPY represents a different mix of economic exposures at the end of the sample than at the beginning. This affects the interpretation of SPY as a control variable but not the internal validity of the pairwise tests among the energy names.

3. EXPLORATORY DATA ANALYSIS

3.1. Descriptive statistics

Table 3.1 reports the moments of the daily log return series. Kurtosis is reported in excess form therefore Gaussian distribution takes the value zero rather than three.

Table 3.1: Descriptive statistics of daily log returns, 2000 to 2026

	COP	CVX	XOM	XLE	SPY
count	6679.000000	6679.000000	6679.000000	6679.000000	6679.000000
mean	0.000411	0.000373	0.000328	0.000326	0.000312
sd	0.020970	0.017289	0.016555	0.018202	0.012156
min	-0.285552	-0.250062	-0.150271	-0.224910	-0.115887
q25	-0.009715	-0.007875	-0.007784	-0.008277	-0.004607
median	0.000627	0.000838	0.000475	0.000808	0.000693
q75	0.011392	0.009072	0.008836	0.009751	0.005971
max	0.224854	0.204904	0.158631	0.152503	0.135577
skew	-0.492665	-0.472797	-0.089883	-0.651333	-0.208995
kurt	12.658779	19.390471	8.351989	12.210782	11.545178

3.1.1. Central tendency

Mean daily returns lie between 0.031% and 0.041% and are economically indistinguishable from one another that frequency. Annualising the mean by a factor of 252 gives a clearer picture and annualising the standard deviation by $\sqrt{252}$ permits a crude risk adjustment.

	Annualised mean	Annualised volatility	Ratio
COP	10.4%	33.3%	0.31
CVX	9.4%	27.4%	0.34
XOM	8.3%	26.3%	0.32
XLE	8.2%	28.9%	0.28
SPY	7.9%	19.3%	0.41

Table 3.2: Annualised moments of daily log returns, 1999 to 2026

Two observations follow. The energy names delivered modestly higher returns than the broad market over the full sample, but did so with substantially greater volatility and on

a simple return to volatility basis the broad market dominates all four energy exposures. Secondly, the tight dispersion of annualised return to volatility ratios among the energy series, $0.28 - 0.34$, indicates that long run performance across these equities is driven by a shared macroeconomic factor rather than a firm specific risk. This therefore supports the existence of a shared long run equilibrium suggesting that short term spread dislocation represents tradeable and mean reverting deviation rather than structural divergence.

3.1.2. Dispersion and Volatility Ordering

The ordering of standard deviations is COP at 2.10%, XLE at 1.82%, CVX at 1.73%, XOM at 1.66% and SPY at 1.22%. This structural ordering aligns with portfolio theory. As a broad market index of 500 equities, SPY diversifies away business risk, retaining exposure solely to systematic risk. The sector ETF (XLE) captures industry wide commodity risk while diversifying away company specific shocks.

Among the single stocks, COP exhibits the highest daily dispersion (2.10%). This elevated volatility is consistent with its structural identity as a pure-play exploration and production (E&P) firm, leaving its equity return profile levered directly to unhedged crude oil price fluctuations. In contrast, the integrated operations of XOM and CVX dampen return variance through downstream refining buffers.

3.1.3. Skewness

All five series exhibit negative skewness, ordered as XLE (-0.65), COP (-0.49), CVX (-0.47), SPY (-0.21) and XOM (-0.09).

The distributional signature of negative skewness is directly whereby for 4 out of 5 assets their median return exceeds the mean return.

This indicates that the left tail of the return distribution is longer and heavier than the right tail. Conditional on a large market move occurring, downside shocks are significantly larger in absolute magnitude than upside rallies. Economically, this means that these assets

appreciate through the steady accumulation of many small daily gains but depreciate through a small number of sharp and severe losses. Notably, XOM exhibits near symmetry at -0.09 .

3.1.4. Kurtosis and the comparison with the original study

Excess kurtosis is the moment of substantive interest, since it measures the weight in the tails relative to a Gaussian benchmark. Values range from 8.35 for XOM to 19.39 for CVX. For context, a Gaussian distribution has excess kurtosis of exactly zero and values above ten indicate that extreme observations occur at a frequency that a normal model does not remotely accommodate.

Comparing against the original study’s estimates over the shorter 2000 to 2018 window is instructive.

	[1], 2000 to 2018	This study, 2000 to 2026
COP	5.47	12.66
CVX	10.64	19.39
XOM	10.48	8.35
XLE	9.07	12.21
SPY	10.45	11.55

Table 3.3: Excess kurtosis: Original against Extended Sample

Kurtosis being a normalised fourth moment means extending a sample by adding several years of ordinary trading days ordinarily dilutes it, because the denominator grows while the extreme observations in the numerator stay fixed. However, 4 of the 5 series rose and CVX nearly doubled. That outcome is only possible if the extended period contains observations extreme enough to overwhelm the dilution, which is a reflection of compounding structural shocks mention in Section 1.

XOM is the exception which is informative. Its excess kurtosis fell from 10.48 to 8.35, the pattern one expects from sample dilution. The explanation is in its extreme values. XOM’s minimum daily return of -0.150271 and maximum of 0.158631 are identical to those reported in the original study over the shorter sample period. Both of XOM’s most extreme

days in the full 26 year sample therefore occurred before January 2019 and 2020 did not set a new record for it. COP, CVX, XLE and SPY all set new extremes in the appended period. Whatever the cause, the largest of the majors absorbed the post 2018 dislocations without registering a daily move outside its existing historical range.

3.1.5. Outliers and the Collapse of Gaussian Tail Model

Dividing each assets series' minimum daily reurn by its standard deviation converts the worst observed day into a standardised z -score. This score enables an empirical test of Gaussian model's validity in describing tail risk.

Table 3.4: Worst observed daily return expressed in standard deviation units (z -scores).

Ticker	Minimum Daily Log Return	Standard Deviations Below Mean (z)
COP	-0.2856	-13.6
CVX	-0.2501	-14.5
XLE	-0.2249	-12.4
SPY	-0.1159	-9.6
XOM	-0.1503	-9.1

These extreme magnitudes are far more than statistical anomalies. The probabilities assigned to them by a Gaussian distribution are so infinitesimally small that they lack any physical real world interpretation.

The Mechanics of Gaussian Tail Decay. Standardising a return R_t under the assumption that returns are normally disributed ($R_t \sim N(\mu, \sigma^2)$) gives the standard score:

$$z = \frac{R_t - \mu}{\sigma}, \quad (3.1)$$

The probability of observing a negative shock (loss) at least as severe as k standard deviations corresponds to the lower tail of the standard normal Cumulative Distribution

Function (CDF):

$$P(Z \leq -k) = \Phi(-k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{-k} e^{-\frac{x^2}{2}} dx \quad (3.2)$$

For extreme tail events where $k \geq 4$, direct numerical integration of Equation (3.2) causes computer to forcibly round to 0.0 a phenomenon known as floating point underflow in standard computational environments. Applying Mills' Ratio [?] bypasses the flaw in numerical integration by evaluating the tail probability via a closed-form asymptotic approximation:

$$P(Z \leq -k) \approx \underbrace{\frac{1}{k\sqrt{2\pi}}}_{\text{algebraic prefactor}} \times \underbrace{e^{-\frac{k^2}{2}}}_{\text{super-exponential decay}}, \quad k \rightarrow \infty. \quad (3.3)$$

Equation (3.3) reveals a weakness of the Gaussian distribution in modeling tail risk. Because k enters the exponent as k^2 , the tail probability decays far faster and the effect of a larger shock compounds instead of merely accumulating.

Worked evaluation: Chevron (*CVX*) at $k = 14.5$. To illustrate the severity of this decay, consider Chevron's minimum daily return, which sits 14.5 standard deviations below its sample mean. Evaluating Equation (3.3) step-by-step:

$$1. \text{ Exponent Evaluation: } \frac{k^2}{2} = \frac{14.5^2}{2} = \frac{210.25}{2} = 105.125 \quad (3.4)$$

$$2. \text{ Exponential Decay Factor: } e^{-105.125} \approx 2.2 \times 10^{-46} \quad (3.5)$$

$$3. \text{ Asymptotic Multiplier: } \frac{1}{k\sqrt{2\pi}} = \frac{1}{14.5 \times 2.5066} \approx 0.0275 \quad (3.6)$$

$$4. \text{ Daily Tail Probability: } P(Z \leq -14.5) \approx 0.0275 \times (2.2 \times 10^{-46}) \approx \mathbf{6 \times 10^{-48}} \quad (3.7)$$

Expected Time. Assuming daily asset returns are independent and identically distributed (i.i.d.), the number of trading days T until a shock of magnitude k occurs follows a discrete

geometric distribution with expected value:

$$E[T] = \frac{1}{p(Z \leq -k)} \quad (3.8)$$

For Chevron's worst day ($k = 14.5$):

$$E[T] \approx \frac{1}{6 \times 10^{-48}} \approx 1.7 \times 10^{47} \text{ trading days} \quad (3.9)$$

Converting trading days into calendar years assuming 252 trading days per year:

$$\text{Expected Years} = \frac{1.7 \times 10^{47}}{252} \approx 6.6 \times 10^{44} \text{ years} \quad (3.10)$$

For context, the age of the observable universe is roughly 1.4×10^{10} years. The Gaussian distribution predicts that a single event like Chevron's worst crash should require 10^{34} universe lifetimes to observe once. Yet, this event appears in a dataset of 6,679 observations (≈ 26 years).

Tail Probabilities Across The Sample. Applying Equation (3.3) across all assets demonstrates that this model breakdown is uniform across market sectors.

	k	$P(Z \leq -k)$ under $N(0,1)$	Implied waiting time (years)
XOM	9.1	4×10^{-20}	1×10^{17}
SPY	9.6	4×10^{-22}	1×10^{19}
XLE	12.4	1×10^{-35}	3×10^{32}
COP	13.6	8×10^{-43}	5×10^{39}
CVX	14.5	6×10^{-48}	7×10^{44}

Table 3.5: Gaussian tail probabilities and implied waiting times for worst observed returns.

Even ExxonMobil (XOM), the mildest outlier in the sample at -9.1σ , carries a theoretical Gaussian return period that exceeds the age of the universe by seven orders of magnitude.

Violations of the Independence Assumption & Volatility Clustering. The expected waiting time framework derived above assumes independent returns across time. In reality, market volatility exhibits strong autocorrelation, a phenomenon known as volatility clustering. When returns alternate between low-volatility (σ_{low}) and high-volatility (σ_{high}) regimes, the return distribution transforms from a single Gaussian to a Gaussian mixture. Mathematically, any variance mixture of normal distributions has excess kurtosis and heavier tails than a homoskedastic Gaussian model.

The Gaussian model therefore fails on two independent fronts. Its theoretical tail decays as $e^{-k^2/2}$, far faster than the heavy tails observed empirically and its assumption of a single variance σ ignores the regime dependent volatility observed in real markets.

Alternative Tail Specification: Heavy-Tailed Power Laws. Financial literature shows that daily stock returns are better described by Power Law distributions, where extreme tail probabilities decay polynomially rather than exponentially [9]:

$$P(|X| > x) \sim C \cdot x^{-\alpha} \quad (3.11)$$

where α represents the tail index (typically estimated between $3 \leq \alpha \leq 4$ for daily US equity returns).

Under a Student's t -distribution with $\nu = 3$ degrees of freedom (which exhibits $\alpha = 3$ asymptotic power-law tail decay), the probability of observing a 14.5σ event is roughly $\sim 10^{-4}$. This corresponds to an expected waiting time of a few decades which is a figure fully consistent with observing it once within a 26-year historical sample.

Comparing the two frameworks:

- Gaussian Model: $P(Z \leq -14.5) \approx 10^{-48} \implies E[T] \approx 10^{44}$ years (Physically impossible)
- Student's t Model ($\nu = 3$): $P(Z \leq -14.5) \approx 10^{-4} \implies E[T] \approx$ Decades (Empirically possible)

Note on Degrees of Freedom (ν). In parametric tail modeling, the parameter ν does not represent sample size degrees of freedom ($n - 1$). Rather, ν serves as a shape parameter determining tail thickness, where the tail index $\alpha = \nu$. As $\nu \rightarrow \infty$, the Student's t -distribution converges to the standard Gaussian model. Setting $\nu \in [3, 5]$ reflects empirical tail estimates for financial asset returns, capturing finite variance while modeling severe kurtosis.

Implications for Practical Risk Management. Traditional risk metrics like Value-at-Risk (VaR) rely heavily on normality assumptions. For instance, a 99% Gaussian VaR threshold corresponds to 2.33σ . While a Gaussian model predicts that losses will exceed this threshold on 1% of trading days, it is systematically incapable of quantifying the magnitude of losses beyond that point, as it assigns the extreme tail region a probability mass that rounds to zero.

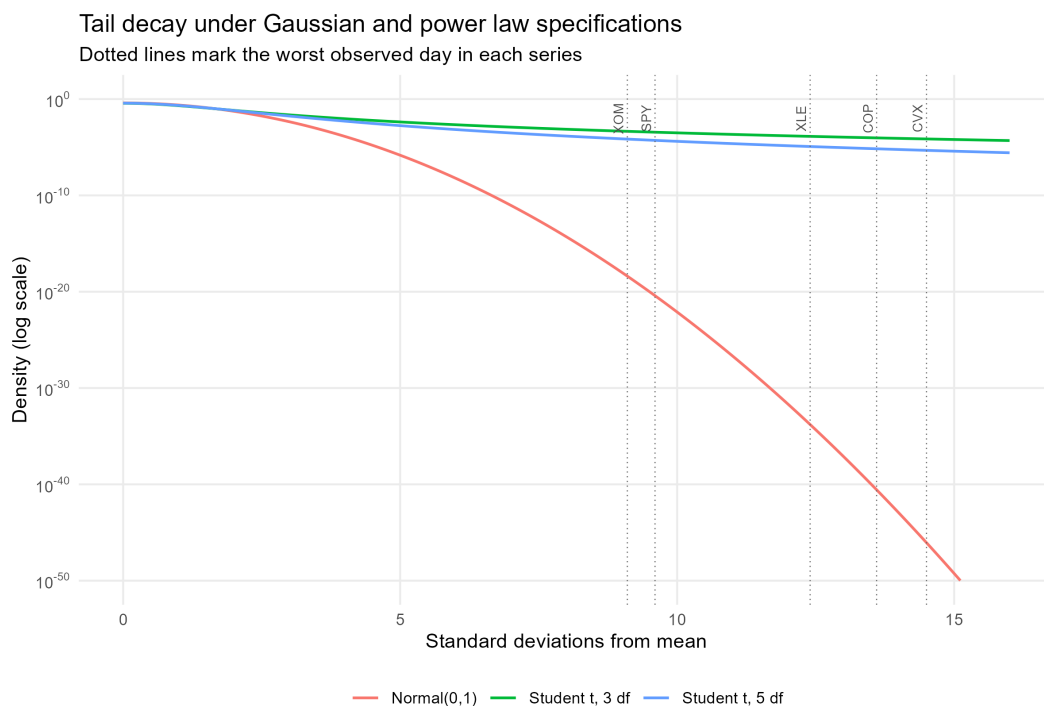


Figure 3.1: Log-density comparison of tail decay under Gaussian vs. heavy-tailed Student's t specifications.

As shown in Figure 3.1, while Gaussian models perform adequately near the distribution

center, they collapse in the tail region where survival is determined.

3.2. Formal tests of normality

To complement the empirical tail analysis in Section 3.1.5, Table 3.6 presents hypothesis tests of the Gaussian null hypothesis ($H_0 : R_t \sim \mathcal{N}(\mu, \sigma^2)$). The Jarque and Bera statistic [10] is a joint omnibus test for skewness and kurtosis together which is reported alongside two other tests that isolate each moment separately, D’Agostino’s test for asymmetry [11] and the Anscombe and Glynn test for tail thickness [12].

Table 3.6: Normality tests on daily log returns					
	Jarque-Bera	p (JB)	p (skew)	p (kurtosis)	Normal?
COP	44,864.9	< 0.0001	< 0.0001	< 0.0001	No
CVX	104,883.8	< 0.0001	< 0.0001	< 0.0001	No
XOM	19,421.4	< 0.0001	0.0027	< 0.0001	No
XLE	41,966.4	< 0.0001	< 0.0001	< 0.0001	No
SPY	37,142.4	< 0.0001	< 0.0001	< 0.0001	No

Every asset’s series rejects normality at all significance levels ($\alpha = 0.01$).

Two features of this results:

Statistical Power against Economic Magnitude. With $N = 6,679$ daily observations, test power is sufficiently high to reject normality over small distributional deviations. Hence the critical insight lies not in the rejection itself, but in the extraordinary scale of the test statistics relative to their critical threshold ($\chi_2^2 = 5.99$ at $\alpha = 0.05$). Exceeding this threshold by three to four orders of magnitude confirms that non-Gaussianity is a dominant structural property rather than an artifact of large sample size.

Asymmetry and Heavy Tails Decomposing the Jarque-Bera statistic into its skewness and kurtosis components reveals a structural divergence across assets. While COP, CVX, XLE and SPY overwhelmingly reject normality across both higher-order moments ($p < 0.0001$),

XOM exhibits relative symmetry ($p = 0.0027$ for skewness). XOM's rejection is driven almost exclusively by kurtosis rather than skewness. This aligns with XOM's near-symmetric raw returns noted and reinforces that tail heaviness due to volatility clustering is the primary mechanism violating normality across the sample.

3.3. Standardised Returns and Graphical Analysis

Each daily log return series is standardised to zero mean and unit variance ($R_t^* \sim (0, 1)$) with a simulated standard normal distribution ($\mathcal{N}(0, 1)$) of equal sample size ($N = 6,679$) is appended as a baseline control. Since standardisation rescales location and dispersion without modifying higher-order moments, the standardised series preserves skewness and kurtosis estimates.

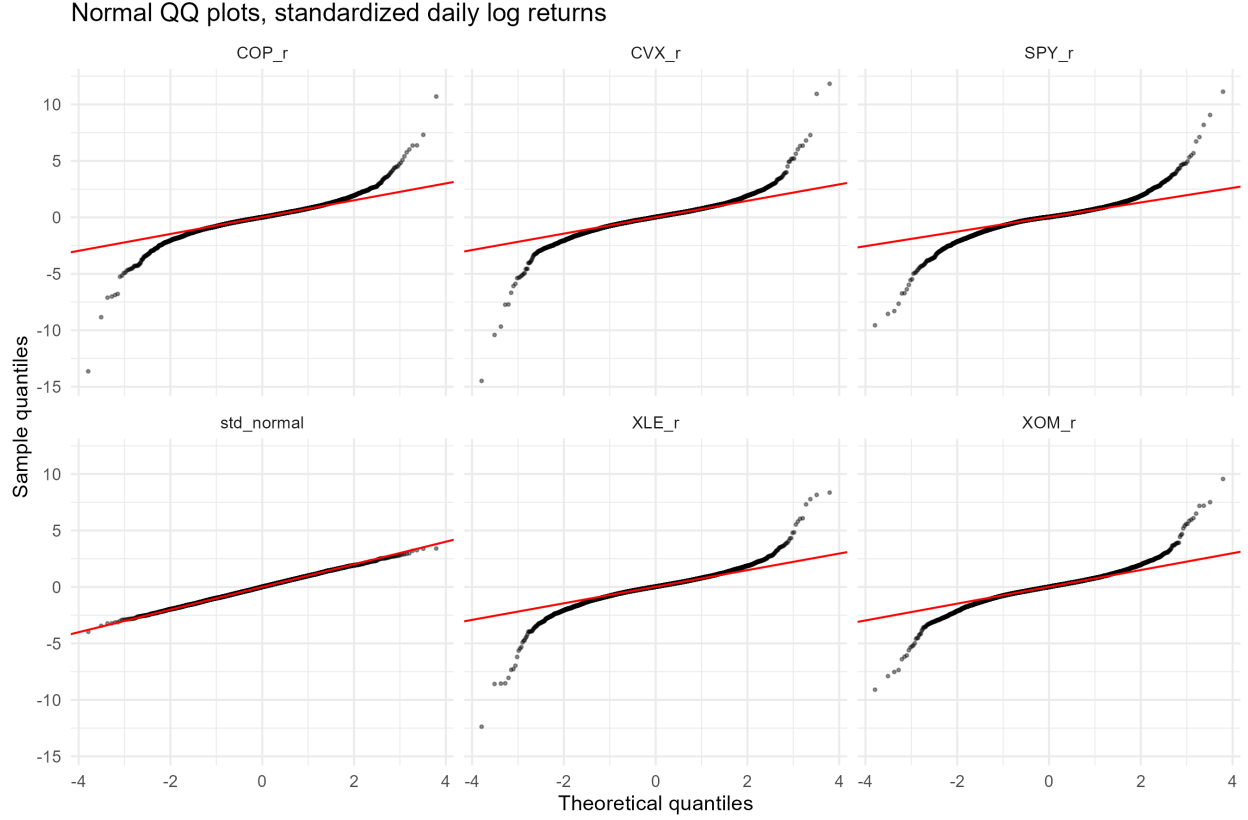


Figure 3.2: Normal quantile-quantile plots of standardised daily log returns against theoretical Gaussian quantiles.

Between approximately -1.5 and $+1.5$ standard deviations every empirical series tracks the 45° reference line closely which means that the Gaussian model describes ordinary trading days adequately. Beyond that range each series departs from the line in a sigmoid like pattern of a heavy-tailed distribution. The departure is also asymmetric as the left tail reaches -10 to -15 standard deviations while the right reaches $+8$ to $+12$ which is consistent with the negative skewness reported in Table 3.1. The simulated normal control tracks the reference line throughout its full range confirming that the departures observed in the simulated series are a property of the data rather than plotting bias. Because the failure is concentrated entirely in the tails, a model calibrated on the well behaved centre of the distribution will systematically understate exactly the risk that determines whether a leveraged or market-

neutral position survives.

3.4. Correlation structure

Table 3.7: Correlation matrix of daily log returns

	COP	CVX	XOM	XLE	SPY
COP	1.0000	0.8110	0.7814	0.8807	0.5520
CVX	0.8110	1.0000	0.8374	0.8931	0.6113
XOM	0.7814	0.8374	1.0000	0.8802	0.5935
XLE	0.8807	0.8931	0.8802	1.0000	0.6592
SPY	0.5520	0.6113	0.5935	0.6592	1.0000

Table 3.7 reports the contemporaneous correlation matrix of daily log returns. The energy sector fund correlates most strongly with each individual major, from 0.78 with XOM to 0.89 with CVX, as expected given that these securities are primary constituents of the fund. Among the individual names, CVX and XOM correlate most strongly with one another at 0.84, consistent with their shared integrated business model, while COP correlates somewhat more weakly with both, at 0.81 and 0.78. SPY is the clear outlier, correlating with the energy names only between 0.55 and 0.66, reflecting exposure to wider economy shocks rather than sector specific shocks.

4. VECTOR AUTOREGRESSION

4.1. Purpose

It is worth stating the function of the Vector Autoregression (VAR) framework within the study because the original study is somewhat ambiguous on the point and would invite a misreading.

The VAR does not generate trading signals, however, it serves a dimensionality reduction device. For five securities, evaluating all relationships involves 10 unordered asset pairs and 20 ordered directional pairs. Subjecting all candidate pairs to full cointegration testing is computationally inefficient for me and would increase the risk of false positives. The VAR therefore acts as a filter to narrow candidate pairs by isolating those with cross predictive information transmission.

As the VAR is estimated on stationary log returns (first differences) as it captures short run dynamics. Although mutual predictability signals local information transmission or a shared transient driver, it does not imply long-run equilibrium co-movement.

4.2. Order of Integration

Two tests are applied. The Augmented Dickey and Fuller (ADF) test takes the presence of a unit root as its null hypothesis, so that the rejection is evidence of stationarity. The Kwiatkowski, Phillips, Schmidt and Shin (KPSS)) test reverses this, taking stationarity as the null, so that rejection is evidence of a unit root. Using both is deliberate. Two tests with opposing nulls that agree constitute a substantially stronger result as the original study relies on the ADF test alone, mine is improved upon here.

Lag lengths for the ADF regressions are selected by the Akaike Information Criterion subject to the upper bound proposed by Schwert,

$$p_{\max} = \left\lfloor 12 \left(\frac{n}{100} \right)^{1/4} \right\rfloor, \quad (4.1)$$

which for $n = 6,680$ gives $p_{\max} = 34$.

Table 4.1: ADF tests on price levels.

series	adf_stat	cv_1pct	cv_5pct	cv_10pct	lags
COP	-0.0755	-3.43	-2.86	-2.57	30
CVX	0.8095	-3.43	-2.86	-2.57	31
XOM	1.5627	-3.43	-2.86	-2.57	34
XLE	0.6419	-3.43	-2.86	-2.57	23
SPY	3.6481	-3.43	-2.86	-2.57	29

Table 4.2: ADF tests on daily log returns.

series	adf_stat	cv_1pct	cv_5pct	cv_10pct	lags
COP	-18.8531	-3.43	-2.86	-2.57	19
CVX	-19.5794	-3.43	-2.86	-2.57	21
XOM	-19.8559	-3.43	-2.86	-2.57	17
XLE	-20.1580	-3.43	-2.86	-2.57	17
SPY	-20.3937	-3.43	-2.86	-2.57	15

Table 4.3: KPSS tests on price levels.

series	kpss_stat	cv_10pct	cv_5pct	cv_2.5pct	cv_1pct
COP	40.636	0.347	0.463	0.574	0.739
CVX	47.447	0.347	0.463	0.574	0.739
XOM	38.017	0.347	0.463	0.574	0.739
XLE	38.987	0.347	0.463	0.574	0.739
SPY	43.634	0.347	0.463	0.574	0.739

The results confirm that all price series are $I(1)$ and $I(0)$ in first differences. As prices, ADF statistics range from -0.08 (COP) to $+3.65$ (SPY), failing to reject the unit root null across all assets. Additionally, KPSS statistics range from 38.0 to 47.4 , rejecting stationarity. Upon taking first differences, ADF statistics range from -18.85 to -20.39 , decisively rejecting the unit root null and confirming stationarity ($I(0)$).

The ADF statistic for the SPY price levels is positive at $+3.65$, which contrasts with ConocoPhillips at -0.08 and may appear anomalous. A positive statistic indicates an estimated autoregressive coefficient marginally above unity, corresponding to a random walk

with positive drift. The fund appreciated from approximately \$92 to \$739 over the sample with a persistent upward drift and this confirms strong nonstationarity driven by a long term economic growth trend.

4.3. VAR Theoretical Framework

4.3.1. Mathematical Formulation of the VAR(p) Model

To capture the linear interdependence among the assets, one needs to formulate a VAR of order p , denoted as VAR(p). Let y_t represent a $K \times 1$ dimensional vector of daily log returns observed at time t , where $K = 5$ corresponds to the the 5 assets in this paper:

$$y_t = \begin{bmatrix} R_{\text{COP},t} \\ R_{\text{CVX},t} \\ R_{\text{XOM},t} \\ R_{\text{XLE},t} \\ R_{\text{SPY},t} \end{bmatrix} \quad (4.2)$$

The general system equation for observation t is expressed as:

$$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \dots + A_p y_{t-p} + e_t \quad (4.3)$$

where c is a $K \times 1$ vector of constant intercept terms:

$$c = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_K \end{bmatrix} \quad (4.4)$$

Each A_l (for $l = 1, \dots, p$) is a $K \times K$ autoregressive coefficient matrix regulating the transmission of returns from lag l to time t :

$$A_l = \begin{bmatrix} a_{1,1}^{(l)} & a_{1,2}^{(l)} & \cdots & a_{1,k}^{(l)} \\ a_{2,1}^{(l)} & a_{2,2}^{(l)} & \cdots & a_{2,k}^{(l)} \\ \vdots & \vdots & \ddots & \vdots \\ a_{K,1}^{(l)} & a_{K,2}^{(l)} & \cdots & a_{K,K}^{(l)} \end{bmatrix} \quad (4.5)$$

The diagonal elements $a_{i,i}^{(l)}$ capture direct own lag return persistence, while the off diagonal elements $a_{i,j}^{(l)}$ ($i \neq j$) capture cross-asset lead lag predictability and spillover shocks across time horizons.

The vector $e_t = [e_{1,t}, e_{2,t}, \dots, e'_{K,t}]$ denotes unobservable $K \times 1$ white noise innovations occurring at time t . The innovation (shock) process satisfies:

$$\mathbb{E}[e_t] = 0 \quad (4.6)$$

$$\mathbb{E}[e_t e'_s] = \begin{cases} \Omega, & \text{if } t = s \\ 0, & \neq s \end{cases} \quad (4.7)$$

where Ω represents the $K \times K$ contemporaneous innovation covariance matrix:

$$\Omega = \mathbb{E}[e_t e'_t] = \begin{bmatrix} \sigma_1^2 & \sigma_{1,2} & \cdots & \sigma_{1,K} \\ \sigma_{2,1} & \sigma_2^2 & \cdots & \sigma_{2,K} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{K,1} & \sigma_{K,2} & \cdots & \sigma_K^2 \end{bmatrix} \quad (4.8)$$

where the diagonal elements $\sigma_i^2 = \text{Var}(e_{i,t})$ denote the unforecastable variance intrinsic to asset i and the off diagonal elements $\sigma_{i,j} = \text{Cov}(e_{i,t}, e_{j,t})$ for $i \neq j$ measure contemporaneous shock co-movements between asset i and asset j occurring within the same trading period t .

Because Ω is a 5×5 symmetric matrix and positive definite, the implied contemporaneous residual correlation between asset innovations is defined as:

$$\rho_{i,j} = \frac{\sigma_{i,j}}{\sigma_i \sigma_j} \quad (4.9)$$

While the coefficient matrices $A_1, \dots, A_p$ capture dynamic intertemporal lead-lag linkages, Ω isolates contemporaneous joint shocks driven by unmodeled sector-wide or macroeconomic news arrivals.

Expanding the system into full matrix notation yields the complete structural representation for observation t :

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \\ \vdots \\ y_{K,t} \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_K \end{bmatrix} = \begin{bmatrix} a_{1,1}^{(1)} & a_{1,2}^{(1)} & \dots & a_{1,K}^{(1)} \\ a_{2,1}^{(1)} & a_{2,2}^{(1)} & \dots & a_{2,K}^{(1)} \\ \vdots & \vdots & \ddots & \vdots \\ a_{K,1}^{(1)} & a_{K,2}^{(1)} & \dots & a_{K,K}^{(1)} \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \\ \vdots \\ y_{K,t-1} \end{bmatrix} + \dots + \begin{bmatrix} a_{1,1}^{(p)} & a_{1,2}^{(p)} & \dots & a_{1,K}^{(p)} \\ a_{2,1}^{(p)} & a_{2,2}^{(p)} & \dots & a_{2,K}^{(p)} \\ \vdots & \vdots & \ddots & \vdots \\ a_{K,1}^{(p)} & a_{K,2}^{(p)} & \dots & a_{K,K}^{(p)} \end{bmatrix} \begin{bmatrix} y_{1,t-p} \\ y_{2,t-p} \\ \vdots \\ y_{K,t-p} \end{bmatrix} + \begin{bmatrix} e_{1,t} \\ e_{2,t} \\ \vdots \\ e_{K,t} \end{bmatrix} \quad (4.10)$$

4.4. Stacked Matrix System Representation

To transition from equation by equation representations to a computationally efficient estimation framework over a sample period of T trading days, we collect daily observation vectors into consolidated block matrices.

Because a $\text{VAR}(p)$ model requires p past observations as historical inputs, to generate its first prediction, the first p time periods cannot be estimated. Consequently, the actual estimation begin at day $t = p + 1$ and spans an effective sample size of $T^* = T - p$ time steps (from $t = p + 1$ through T). The entire multi equation time series system is compactly expressed as:

$$Y = BZ + U \quad (4.11)$$

where Y , B , Z and U are partitioned matrices structured to align temporal and cross

sectional dimensions:

1. **Dependent Variable Matrix (Y):** The matrix $Y \in \mathbb{R}^{K \times (T-p)}$ collects observed daily return vectors side by side across time steps $t = p + 1$ through T :

$$Y = \begin{bmatrix} | & | & & | \\ y_{p+1} & y_{p+2} & \cdots & y_T \\ | & | & & | \end{bmatrix} = \begin{bmatrix} y_{1,p+1} & y_{1,p+2} & \cdots & y_{1,T} \\ y_{2,p+1} & y_{2,p+2} & \cdots & y_{2,T} \\ \vdots & \vdots & \ddots & \vdots \\ y_{K,p+1} & y_{K,p+2} & \cdots & y_{K,T} \end{bmatrix} \quad (4.12)$$

2. **Parameter Matrix (B):** The parameter matrix $B \in \mathbb{R}^{k \times (1+Kp)}$ horizontally concatenates the constant intercept vector c and the sequence of p autoregressive coefficient matrices $[c \ A_1 \ \dots \ A_p]$:

$$B = \begin{bmatrix} c_1 & a_{1,1}^{(1)} & \cdots & a_{1,K}^{(1)} & \cdots & a_{1,1}^{(p)} & \cdots & a_{1,K}^{(p)} \\ c_2 & a_{2,1}^{(1)} & \cdots & a_{2,K}^{(1)} & \cdots & a_{2,1}^{(p)} & \cdots & a_{2,K}^{(p)} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ c_K & a_{K,1}^{(1)} & \cdots & a_{K,K}^{(1)} & \cdots & a_{K,1}^{(p)} & \cdots & a_{K,K}^{(p)} \end{bmatrix} \quad (4.13)$$

3. **Regressor Matrix (Z):** The regressor matrix $Z \in \mathbb{R}^{(1+Kp) \times (T-p)}$ aggregates the explanatory variables. Column t of Z contains the constant intercept unit alongside all lagged return vectors required to predict column t of Y :

$$Z = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ y_p & y_{p+1} & \cdots & y_{T-1} \\ y_{p-1} & y_p & \cdots & y_{T-2} \\ \vdots & \vdots & \ddots & \vdots \\ y_1 & y_2 & \cdots & y_{T-p+1} \end{bmatrix} \quad (4.14)$$

4. **Residual Matrix (U):** The matrix $U \in \mathbb{R}^{K \times (T-p)}$ collects unobserved innovation vectors e_t for each estimated time step:

$$U = \begin{bmatrix} | & | & & | \\ e_{p+1} & e_{p+2} & \cdots & e_T \\ | & | & & | \end{bmatrix} = \begin{bmatrix} e_{1,p+1} & e_{1,p+2} & \cdots & e_{1,T} \\ \vdots & \vdots & \ddots & \vdots \\ e_{K,p+1} & e_{K,p+2} & \cdots & e_{K,T} \end{bmatrix} \quad (4.15)$$

4.5. Closed Form System Estimation via OLS

Every individual equation in the VAR(p) system features an identical design matrix Z on the right-hand side. Because the explanatory variables are identical across all K assets, the Seemingly Unrelated Regression (SUR) system estimator collapses directly into equation by equation Ordinary Least Squares (OLS) without loss of asymptotic efficiency.

To solve for the parameter matrix B , we post-multiply the stacked system $Y = BZ + U$ by the transpose of the regressor matrix, Z' :

$$YZ' = B(ZZ') + UZ' \quad (4.16)$$

The matrix inner product $(ZZ') \in \mathbb{R}^{(1+Kp) \times (1+Kp)}$ transforms the rectangular regressor array into a symmetric, square sample autocovariance matrix. Under the standard Gauss-Markov exogeneity assumption ($\mathbb{E}[UZ'] = 0$), therefore the expected cross-product of lagged regressors and unforecastable innovations cancels out. Dropping the residual term leaves the system of Normal Equations:

$$YZ' = B(ZZ') \quad (4.17)$$

Multiplying both sides by the matrix inverse $(ZZ')^{-1}$ isolates the closed-form OLS parameter estimator $\hat{B}$:

$$\hat{B} = YZ'(ZZ')^{-1} \quad (4.18)$$

This closed-form result means that estimating a system of K equations containing $K(1 + Kp)$ total parameters requires only a single matrix inversion of dimension $(1 + Kp) \times (1 + Kp)$, bypassing iterative optimization routines.

4.6. Residual Extraction and Innovation Covariance Estimation

Once $\hat{B}$ is estimated, we evaluate fitted values $\hat{Y} = \hat{B}Z$ and extract the empirical sample residual matrix $\hat{U}$:

$$\hat{U} = Y - \hat{B}Z \quad (4.19)$$

Column t of $\hat{U}$ yields the estimated $K \times 1$ innovation vector $\hat{e}_t = [\hat{e}_{1,t}, \dots, \hat{e}_{K,t}]'$.

To estimate the contemporaneous innovation covariance matrix $\Omega = \mathbb{E}[e_t e_t']$, we compute the cross product matrix of empirical residuals, $\hat{U}\hat{U}'$:

$$\hat{U}\hat{U}' = \sum_{t=p+1}^T \hat{e}_t \hat{e}_t' \quad (4.20)$$

To ensure that $\hat{\Omega}$ is an unbiased estimator, we adjust the scale factor for loss of degrees of freedom. The effective sample size is $T^* = T - p$, and each equation in the VAR system consumes $1 + Kp$ degrees of freedom to estimate its parameter vector. The unbiased estimator for the $K \times K$ innovation covariance matrix is defined as:

$$\hat{\Omega} = \frac{1}{(T - p) - (1 + Kp)} \hat{U}\hat{U}' \quad (4.21)$$

4.7. Lag order selection

Determining the autoregressive lag order p requires balancing model fit against overparameterisation. Choosing an excessively large p risks overfitting in sample noise and inflating estimator variance. On the other hand selecting a lag order that is too small introduces omitted variable bias.

The lag order p is formally selected by minimising a Information Criterion $IC(p)$ evaluated across a candidate grid $p \in \{1, 2, \dots, p_{\max}\}$. Each criterion balances a goodness-of-fit metric which is measured by the log determinant of the estimated innovation covariance matrix $\ln |\hat{\Omega}(p)|$ against a penalty term that increases with the total number of system parameters pK^2 :

$$\text{AIC}(p) = \ln |\hat{\Omega}(p)| + \frac{2}{T}pK^2, \quad (4.22)$$

$$\text{HQ}(p) = \ln |\hat{\Omega}(p)| + \frac{2 \ln \ln T}{T}pK^2, \quad (4.23)$$

$$\text{SC}(p) = \ln |\hat{\Omega}(p)| + \frac{\ln T}{T}pK^2. \quad (4.24)$$

Akaike's Final Prediction Error (FPE) criterion, which is asymptotically equivalent to the AIC in moderate and large samples, is evaluated alongside AIC for completeness:

$$\text{FPE}(p) = \left(\frac{T + Kp + 1}{T - Kp - 1} \right)^K |\hat{\Omega}(p)| \quad (4.25)$$

When evaluated on the empirical sample ($T = 6,677$, $K = 5$) across a maximum lag search space of $p_{\max} = 20$, the criteria yield conflicting choices, as reported in Table 4.4.

	AIC	HQ	SC	FPE
Selected order	9	1	1	9

Table 4.4: Lag order selected by each Information Criterion, $p_{\max} = 20$

At $T = 6,677$, the Schwarz (SC/BIC) penalty carries a factor of $\ln T \approx 8.81$ per parameter,

compared to the fixed Akaike (AIC) factor of 2. SC/BIC is therefore facing a tax ratio exceeding four to one. Because each additional lag introduces $K^2 = 25$ new coefficients across the system, the two criteria price model expansion at fundamentally different rates. The Hannan-Quinn (HQ) and Schwarz (SC) criteria are consistent estimators that converge to the true lag order as $T \rightarrow \infty$ whereas AIC and FPE are known to over select asymptotically in large samples.

4.7.1. Mathematical Inspection of AIC

To explicitly demonstrate the marginal nature of the AIC preference for $p = 9$, we evaluate the criterion values across $p = 1$ and $p = 9$. Given $T^* = 6,677$ and $K = 5$ ($K^2 = 25$ parameters per lag), the criterion values evaluate as follows:

$$\text{AIC}(1) = \ln |\hat{\Omega}(1)| + \frac{2}{6,677}(25) = -46.1791 \quad (4.26)$$

$$\text{AIC}(9) = \ln |\hat{\Omega}(9)| + \frac{2}{6,677}(225) = -46.1977 \quad (4.27)$$

The absolute reduction in the criterion metric is:

$$\Delta\text{AIC} = 46.1977 - 46.1791 = 0.0186 \quad (4.28)$$

Scaling this reduction against the base metric at $p = 1$ yields the relative gain:

$$\text{Relative Gain} = \frac{0.0186}{46.1791} \approx 0.0004027 \quad \left(\text{or } \approx \frac{4}{10,000} \right) \quad (4.29)$$

Estimating 200 additional parameters yields a 0.04% relative improvement in fit, confirming that $p = 9$ captures sampling noise rather than macroeconomic structure.

As illustrated in Figure 4.1, the smoothed AIC surface displays an essentially flat profile beyond the initial lags. Although the metric reaches a technical global minimum at $p = 9$,

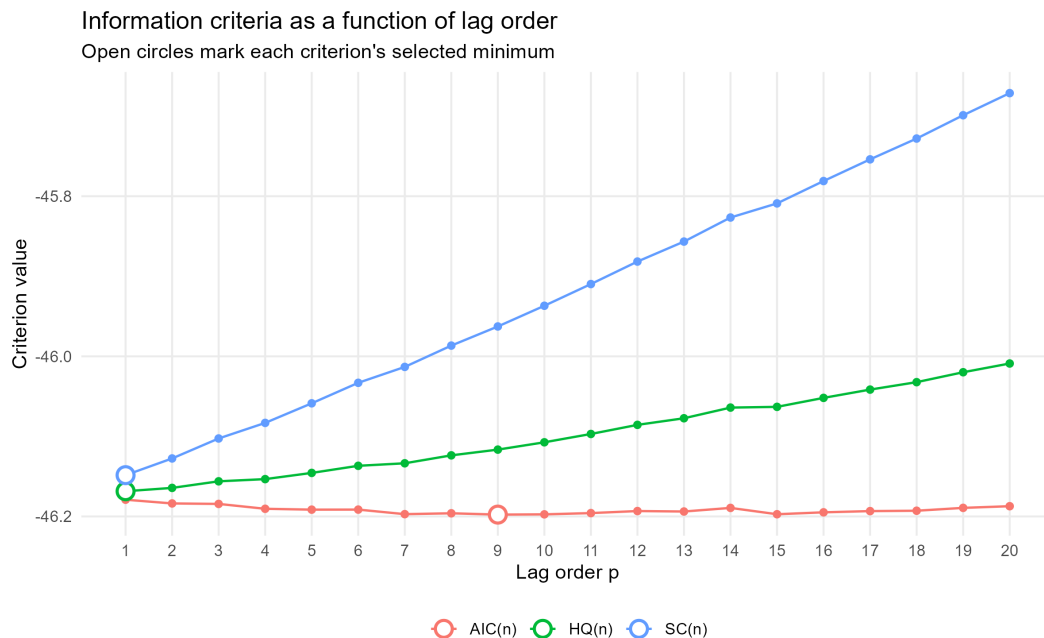


Figure 4.1: Information Criteria evaluated across VAR lag orders

the curve reflects a near-zero slope across the entire candidate space $p \in [1, 10]$. This almost smooth horizontal plateau confirms that incremental lags beyond $p = 1$ yield negligible information gains, causing AIC to slowly drift toward a shallow minimum driven by high-frequency sample variation rather than strong dynamic feedback. By contrast, the Schwarz criterion increases monotonically for all $p > 1$, reflecting an unambiguous preference for model parsimony.

Neither criterion selects $p = 2$, which was the order specified in the baseline study. This paper maintains $p = 2$ as the primary specification purely to ensure direct comparability with the benchmark literature. However, acknowledging that statistical criteria over this extended sample favor $p = 1$.

4.8. Estimation Results

Table 4.5 reports the estimated coefficients of the VAR(2) system fitted to 6,677 observations across the 5 assets

Table 4.5: VAR(2) Coefficient Estimates and Model Fit Statistics

	COP	CVX	XOM	XLE	SPY
COP.L1	-0.0434.	-0.0334	-0.0447*	-0.0051	-0.0188
CVX.L1	-0.1447***	-0.1249***	-0.0826**	-0.1087***	-0.0514**
XOM.L1	0.0428	0.0173	0.0110	0.0462	0.0107
XLE.L1	0.1153*	0.1150**	0.0856*	0.0257	0.0249
SPY.L1	-0.0703*	-0.0942***	-0.1077***	-0.0259	-0.0572***
COP.L2	-0.0094	0.0197	0.0046	0.0119	0.0123
CVX.L2	-0.0154	-0.0019	0.0063	-0.0197	0.0148
XOM.L2	-0.0669*	-0.0814**	-0.0998***	-0.0785**	-0.0365.
XLE.L2	0.0353	0.0210	0.0251	0.0246	0.0137
SPY.L2	0.0618*	0.0477*	0.0260	0.0404	-0.0354*
const	0.0005.	0.0004*	0.0004*	0.0004.	0.0004*
R^2	0.0078	0.0116	0.0148	0.0067	0.0111
Adj. R^2	0.0063	0.0101	0.0133	0.0052	0.0096
F statistic	5.21	7.82	10.02	4.48	7.49
Resid. s.e.	0.02090	0.01720	0.01644	0.01815	0.01209

Notes: Dependent variables are daily returns. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, $p < 0.10$.

4.8.1. Goodness of Fit

Coefficients of determination (R^2) range from 0.0067 for XLE,Md to 0.0148 for Exxon Mobil. This means that this system explains $< 1\frac{1}{2}\%$ of the variation in daily returns in every equation. The F -statistics significant at all conventional level, ranging from 4.48 to 10.02 with p -values below 10^{-5} throughout. There is therefore genuine predictability in the statistical sense.

4.8.2. Stability

A VAR(p) system is dynamically stable if all eigenvalues of its companion matrix lie strictly within the complex unit circle i.e if the roots of the characteristic polynomial $|\lambda I - C| = 0$ satisfy $|\lambda| < 1$.

To formulate this, consider a K variable VAR(p) process,

$$Y_t = A_1 Y_{t-1} + A_2 Y_{t-2} + \cdots + A_p Y_{t-p} + u_t. \quad (4.30)$$

To evaluate dynamic stability, the p lag system is mapped into an equivalent Kp dimensional VAR(1) companion form,

$$Z_t = CZ_{t-1} + U_t, \quad (4.31)$$

where

$$Z_t \equiv \begin{bmatrix} Y_t \\ Y_{t-1} \\ \vdots \\ Y_{t-p+1} \end{bmatrix}, \quad U_t \equiv \begin{bmatrix} u_t \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad (4.32)$$

and C is the $Kp \times Kp$ companion matrix

$$C = \begin{bmatrix} A_1 & A_2 & \cdots & A_{p-1} & A_p \\ I_K & 0 & \cdots & 0 & 0 \\ 0 & I_K & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & I_K & 0 \end{bmatrix}. \quad (4.33)$$

For a 5-variable VAR(2) model ($K = 5, p = 2$), C is a 10×10 matrix:

$$C = \begin{bmatrix} A_1 & A_2 \\ I_5 & 0_5 \end{bmatrix} \quad (4.34)$$

The computed root moduli for this system are given by:

$$\{|\lambda_i|\}_{i=1}^{10} = \{0.256, 0.256, 0.243, 0.243, 0.167, 0.167, 0.144, 0.144, 0.135, 0.135\}. \quad (4.35)$$

All roots lie comfortably within the unit circle, with the maximum root modulus bounded at $\max_i |\lambda_i| \approx 0.256 < 1$. This satisfies the necessary and sufficient conditions for stationarity and dynamic stability.

The appearance of roots in exact duplicate pairs indicates complex conjugate eigenvalues, $\lambda_i = a \pm bi$. Economically, this implies that systemic decay follows an oscillatory path rather than a monotonic dampening. Furthermore, the low maximum eigenvalue modulus ($\max|\lambda| \approx 0.256$) shows a very weak shock persistence in the VAR system. Dynamic response functions decay at rate $(\max|\lambda|)^k$, a unit disturbance dissipates by over 93% within two trading days ($0.256^2 \approx 0.065$), leaving under 6.6% residual impact. This rapid decay then confirms that cross asset information is absorbed almost immediately making the system return to equilibrium within a short time.

4.8.3. Coefficient Estimates Interpretation

Two economic patterns emerge from the coefficient signs.

Firstly is the short run mean reversion by the negative coefficients. First lag coefficients for CVX CVX.L1 and SPY SPY.L1 are uniformly negative and statistically significant. For instance a 1% positive return in CVX predicts a -0.145% return in COP ($p < 0.001$) and a -0.109% return in XLE ($p < 0.001$) on the subsequent trading day. This negative sign pattern in daily high frequency markets tend to be due to market microstructure features such as profit taking, bid-ask bounce and market marker rebalancing.

Another observation is in the sector momentum transmission seen in the positive coefficients. Lagged XLE returns (XLE.L1) yield consistently positive coefficients for COP ($+0.1153, p < 0.05$), CVX ($+0.1150, p < 0.01$), and XOM ($+0.0856, p < 0.05$). This highlights an asymmetric transmission mechanism whereby individual firm shocks trigger immediate mean reverting corrections while aggregate sector ETF flows generate positive short term momentum across member stocks.

4.8.4. Innovation Covariance Structure

Contemporaneous residual correlations among the three integrated majors and the sector fund are high, between 0.78 and 0.89, with the sector fund most strongly correlated with

Table 4.6: Contemporaneous correlation matrix of VAR(2) residuals.

	COP	CVX	XOM	XLE	SPY
COP	1.0000	0.8097	0.7805	0.8804	0.5506
CVX	0.8097	1.0000	0.8363	0.8937	0.6095
XOM	0.7805	0.8363	1.0000	0.8808	0.5899
XLE	0.8804	0.8937	0.8808	1.0000	0.6588
SPY	0.5506	0.6095	0.5899	0.6588	1.0000

each individual asset. Correlations involving the broad market fund are markedly lower, between 0.55 and 0.66.

Two implications follow. First, the broad market fund is distinguishable from the four energy exposures in a way that supports its treatment as a control asset rather than as a candidate pair leg. Second, the presence of high contemporaneous residual correlations ($r \approx 0.89$) among the energy series highlights an identification problem in reduced-form VAR analysis. Because reduced-form innovations are non-orthogonal, structural Impulse Response Function (IRF)s derived via lower-triangular Cholesky factorization rely heavily on the imposed recursive causal ordering. When residual covariance is high, assigning common contemporaneous shocks to a specific variable becomes mathematically arbitrary. The Cholesky decomposition is nonetheless applied in order to demonstrate this sensitivity directly.

A comparison with the original study highlights how sample extension across volatile macro regimes alters error covariance structure. In the original sample, the contemporaneous residual correlations between the sector benchmark (XLE) and the individual majors (COP, CVX and XOM) were recorded at 0.855, 0.878 and 0.854, respectively. In our extended sample, these parameters shift upward to 0.880, 0.894 and 0.881.

This spike in residual co movement aligns with the property of asymmetric cross asset correlation during systemic downturns. As established in Section 3.1.5, the extended sample incorporates the unprecedented energy market volatility of 2020, a period characterized by simultaneous crude demand destruction and forced institutional portfolio deleveraging.

During such tail events, asset prices cease to trade on company specific fundamentals and instead respond synchronously to macro level shocks. Since covariance calculations are heavily weighted by extreme realizations, these clustered outliers pull the overall linear dependency upward, rendering the identification problem even more acute in the post-2020 sample than in the original benchmark.

4.8.5. Impulse Response Analysis

4.8.5.1. Identification and ordering problem Because the $VAR(2)$ system is stable with all characteristic roots being below 0.26 in modulus, it therefore admits that an infinite order vector representation

$$Y_t = \sum_{s=0}^{\infty} \Phi_s u_{t-s}, \quad (4.36)$$

in which the matrix Φ_s records the response of the system at horizon s to a unit disturbance in each system. A disturbance to equation 4.36 occurs alongside other disturbances simultaneously, so Φ_s contradicts the response of interest with contemporaneous co-movements. The conventional solution is constructing mutually uncorrelated shocks w_t satisfying $w_t w_t'$ through the factorisation

$$\Omega = PP', \quad u_t = Pw_t, \Theta_s = \Phi_s P, \quad (4.37)$$

where P is the unique lower triangular Cholesky factor of Ω with a positive diagonal, reporting the orthogonalised response Θ_s in place of Φ_s . The triangularity of P is an identifying

restriction imposed by the researcher rather than a property recovered from the data:

$$P = \begin{bmatrix} p_{11} & 0 & 0 & \cdots & 0 \\ p_{21} & p_{22} & 0 & \cdots & 0 \\ p_{31} & p_{32} & p_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ p_{K1} & p_{K2} & p_{K3} & \cdots & p_{KK} \end{bmatrix}. \quad (4.38)$$

The upper triangular zeros encode a strict recursive contemporaneous ordering where the first variable affects every other variable within the period and affect by none while the last variable is affected by every other and affects none.

4.8.5.2. *Ordering specifications*

Ordering A (Primary) Runs from most to least aggregated:

$$\text{SPY} \rightarrow \text{XLE} \rightarrow \text{XOM} \rightarrow \text{CVX} \rightarrow \text{COP}$$

This is based on the reasoning that a market wide disturbance is transmitted to individual constituents within a single session more readily than a firm specific disturbance is transmitted to a broad index. This is a claim about the speed and direction of transmission at daily frequency and carries no implication about the relative magnitude or economic importance of systematic and non-systematic shocks.

Ordering B (Reversal) The exact inversion:

$$\text{COP} \rightarrow \text{CVX} \rightarrow \text{XOM} \rightarrow \text{XLE} \rightarrow \text{SPY}$$

This serves as a diagnostic based on unfalsifiability: an assumption untestable against the data directly is instead tested against its own negation.

Confidence bands across all specifications are computed from 500 bootstrap replications at the 95% confidence level over a 6 period forecast horizon.

4.8.6. Marginal Response

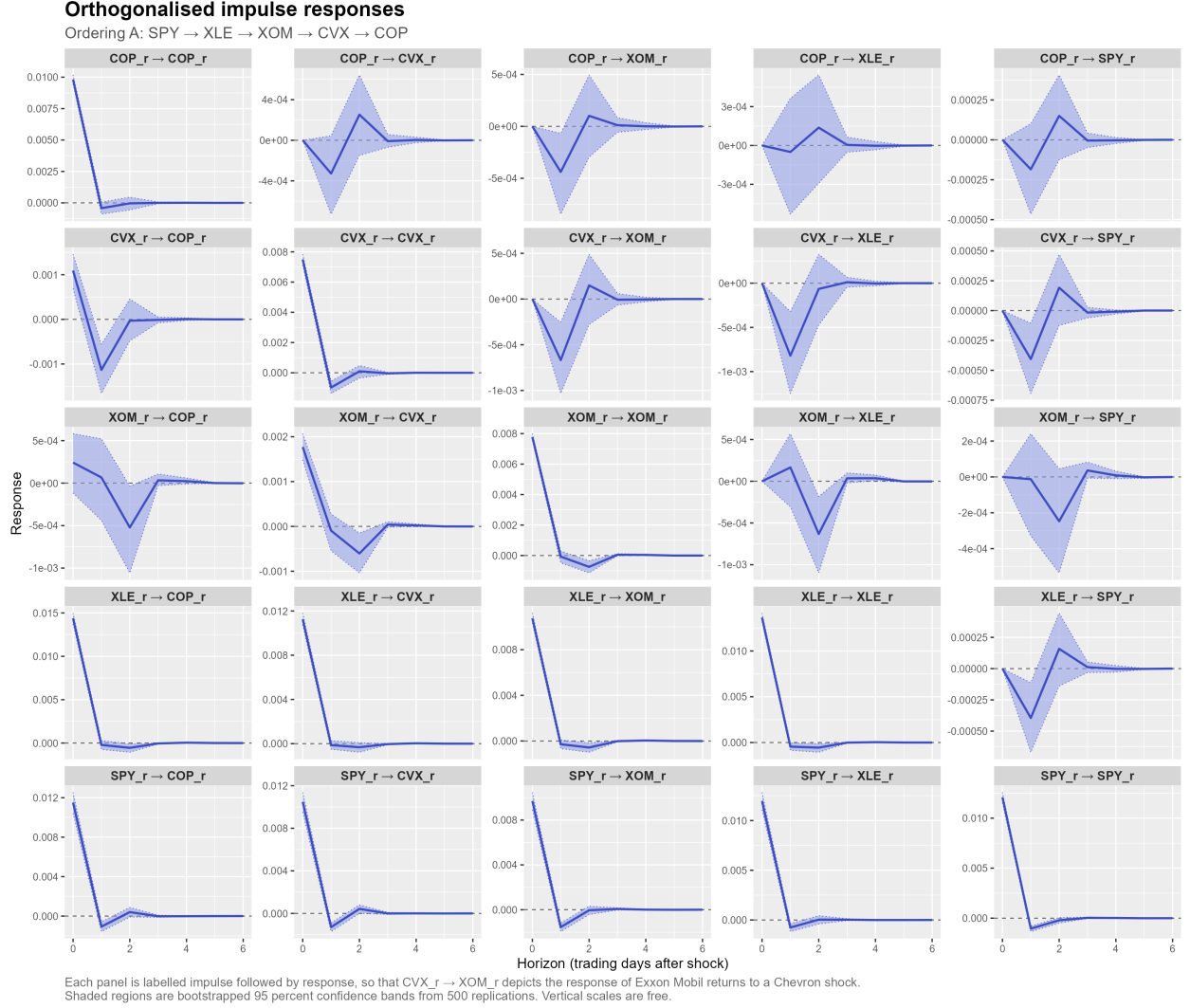


Figure 4.2: Orthogonalised impulse responses under Ordering A. Each panel is labelled impulse followed by response, so that CVX_r $\rightarrow$ XOM_r depicts the response of Exxon Mobil returns to a Chevron shock. Shaded regions are bootstrapped 95 percent confidence bands.

The shocks dissipate rapidly. The largest response anywhere in the system falls by approximately 90% within a single trading day and by 95% within two and is numerically indistinguishable from zero by the seventh. This is consistent with the maximum characteristic root of 0.256 and matches the qualitative finding of the original study, which likewise reports a pronounced fade away effect after the initial impulse.

The decay is frequently oscillatory rather than monotone. Many responses overshoot into negative territory at horizon one before returning towards zero from below. This reproduces at the system level the negative first order coefficients and is the expected signature of the complex conjugate eigenvalues reported.

4.8.7. Cumulative Response

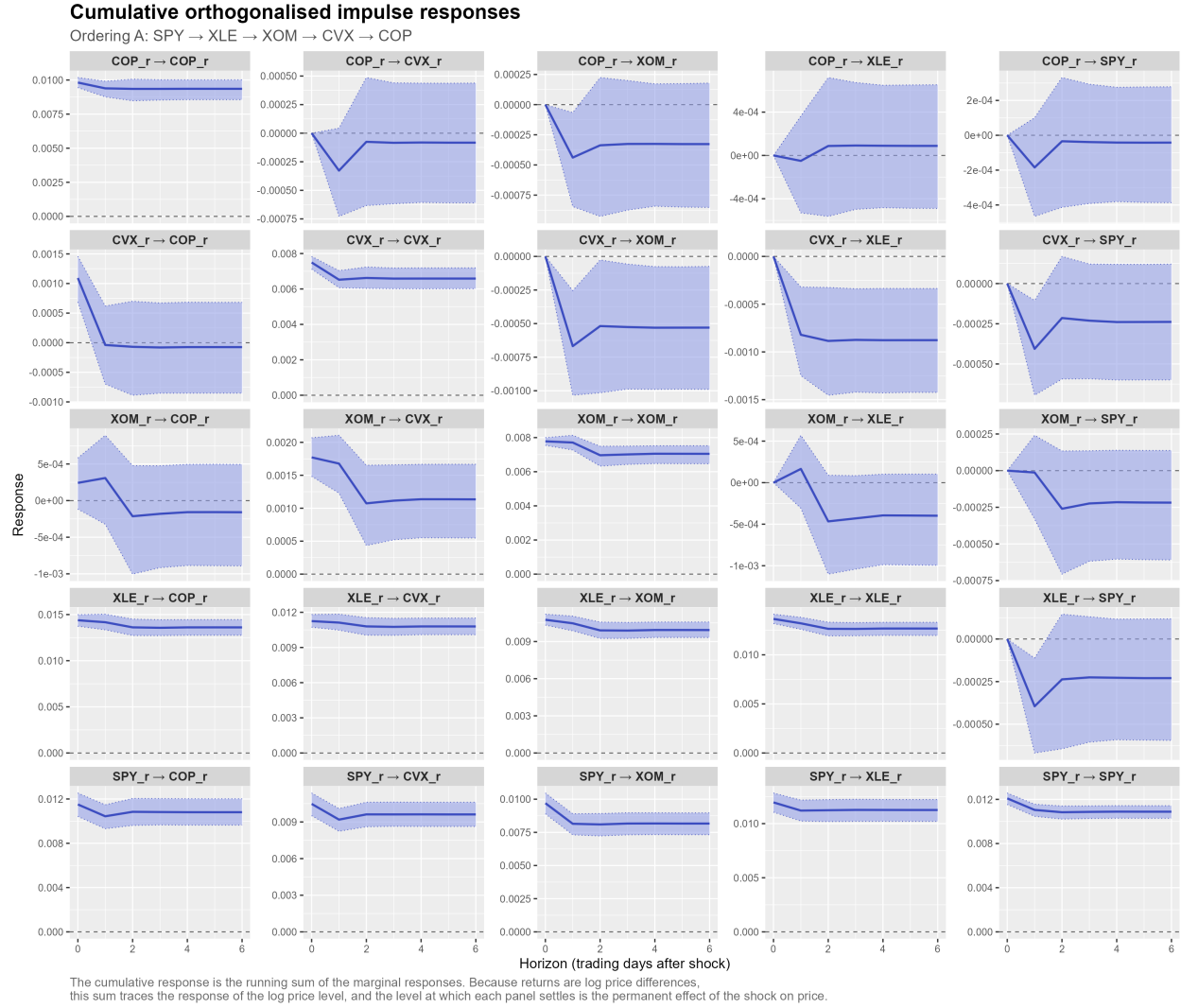


Figure 4.3: Cumulative orthogonalised impulse responses under Ordering A. Each panel traces the response of the log price level and the level at which it settles is the permanent effect of the shock.

Every own shock cumulative response converges to a nonzero constant rather than returning to the origin and Table 4.7 reports the levels at which they settle. The permanent effect of a one standard deviation own shock is approximately 0.94% of the log price level for ConocoPhillips, 0.66% for Chevron and 0.71% percent for XOM.

Table 4.7: Permanent impulse response of log prices to an own shock (Ordering A).

Series	Permanent effect	Lower 95%	Upper 95%	Percent
XLE	0.01268	0.01197	0.01331	1.27
SPY	0.01088	0.01029	0.01143	1.09
COP	0.00935	0.00856	0.01001	0.94
XOM	0.00705	0.00647	0.00752	0.70
CVX	0.00659	0.00601	0.00719	0.66

The speed of absorption is more striking than its magnitude. The impact responses under the same ordering are 0.00981, 0.00750 and 0.00778 respectively, so the day one displacement already accounts for 95% of the permanent effect for ConocoPhillips, 88% for Chevron and 91% for Exxon Mobil.

4.8.8. Cholesky Ordering Sensitivity

To assess the sensitivity of the orthogonalised responses to the identification scheme, the primary specification is compared against its exact reversal.

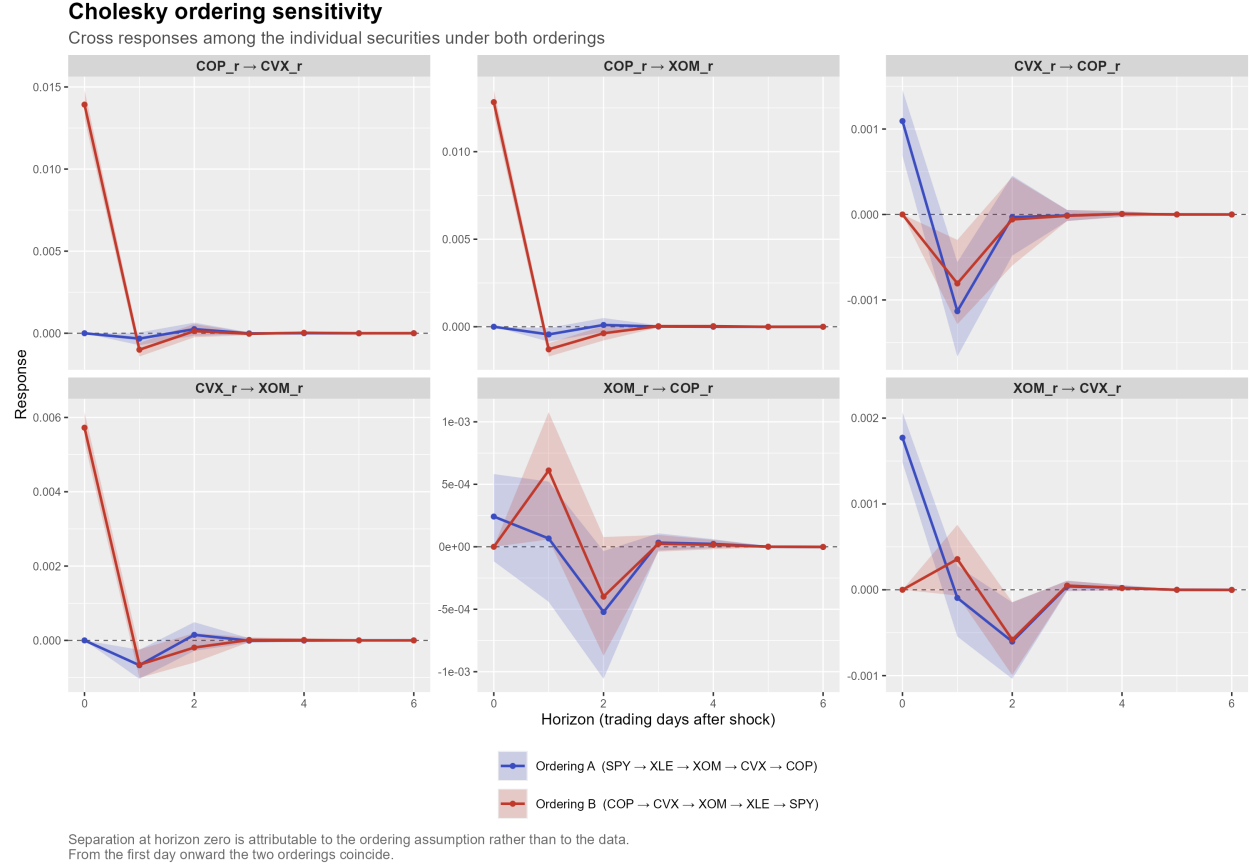


Figure 4.4: Cross responses among the three individual securities under both orderings. Separation at horizon zero is attributable to the ordering assumption rather than to the data. From the first day onward the two orderings coincide.

4.8.8.1. Contemporaneous Instability At $h = 0$, the divergence is total. Every one of the six ordered pairs among the individual securities registers exactly zero under one ordering and a nonzero value under the other. There is no pair in which the two orderings even partially agree.

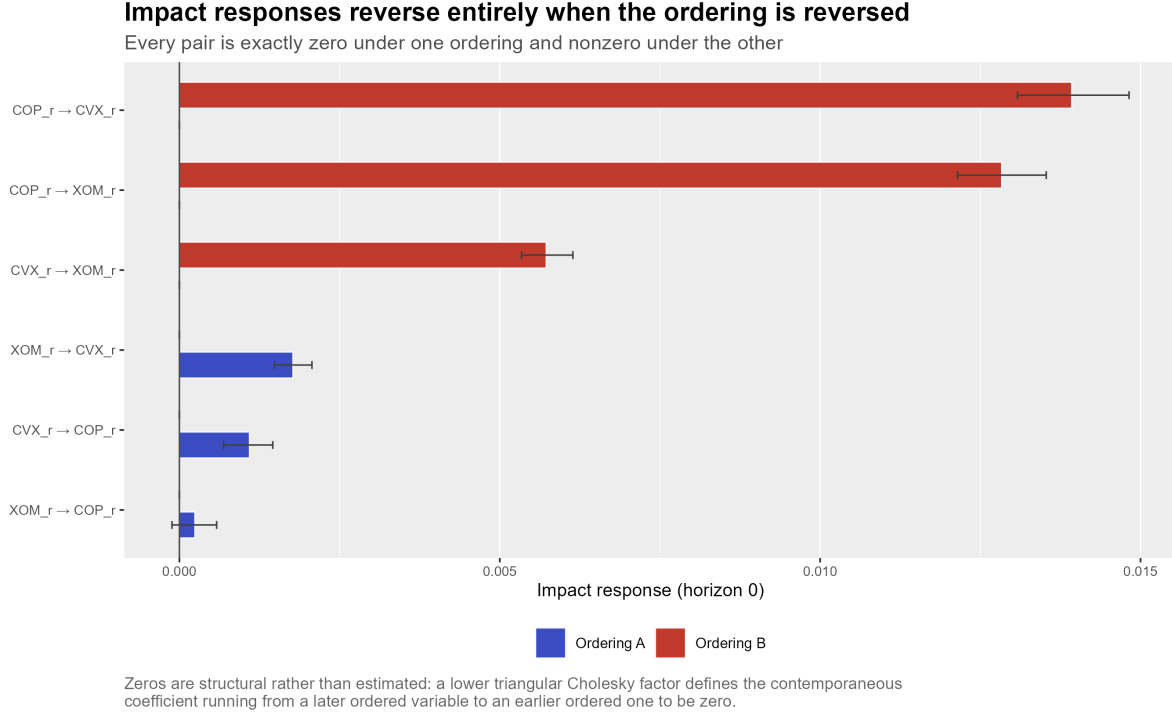


Figure 4.5: Impact responses for the six ordered pairs under both orderings, with bootstrapped 95 percent intervals.

The zeros are structural rather than estimated. Since $\Theta_0 = P$ and P is lower triangular, the contemporaneous coefficient running from a later ordered variable to an earlier ordered one is set to zero by the form of the factorisation. Ordering A places COP last, so its impact responses on CVX and XOM are identically zero. Ordering B places it first and the same responses become 0.01393 and 0.01283.

4.8.8.2. Interpretation Impact responses are not used as structural evidence anywhere in this paper. The cumulative responses require a narrower qualification as their magnitudes and signs inherit the arbitrariness of the impact term, since the cumulative response of CVX to a COP shock is -0.000083 under one ordering and $+0.013042$ under the other but the qualitative conclusion that shocks displace prices permanently holds under both.

Pair selection therefore proceeds on the Granger causality which are computed from lagged coefficient matrices and are invariant to the contemporaneous ordering. That invariance is

the reason those tests rather than these responses serve as the screening device.

4.8.9. Granger Causality

4.8.9.1. Purpose Section 4.8.8 establishes that the orthogonalised impulse responses cannot identify contemporaneous transmission, as reversing the Cholesky ordering flips every impact response in the sample. Screening candidate trading pairs therefore requires a statistic that is strictly invariant to variable sequencing.

Granger causality satisfies this criterion. The test evaluates whether lagged returns of causing series x contains incremental predictive information about a caused series y beyond y 's own historical trajectory since the test relies on the lag coefficient matrices $A_1, \dots, A_p$. Those matrices are then estimated by Ordinary Least Squares (OLS) and the results are completely independent of any contemporaneous factoring applied to the innovation covariance matrix Ω . A permutation of the variables relabels the equations and leaves every coefficient, standard error and test statistic unchanged.

For the ordered pair (y, x) the null hypothesis is that all coefficients on the lags of x in the equation for y are jointly zero,

$$H_0 : \alpha_1 = \alpha_2 = \dots = \alpha_p = 0 \quad \text{in} \quad y_t = c + \sum_{i=1}^p \beta_i y_{t-i} + \sum_{i=1}^p \alpha_i x_{t-i} + \varepsilon_t, \quad (4.39)$$

tested by the standard F statistic on the restriction. Rejection indicates that x Granger causes y , which is a statement about predictive content rather than about causation in any structural sense.

4.8.9.2. Multiple comparisons Evaluating 20 simultaneous hypotheses at a nominal significance level of $\alpha = 0.10$ yields an expected rate of two false rejections under the null hypothesis ($20 \times 0.10 = 2$). The original study reports the pairs surviving at that level with no adjustment, which overstates the evidence by construction.

To resolve this, we report two error corrections. The Benjamini and Hochberg (BH)

procedure controls the False Discovery Rate (FDR), the expected proportion of rejections that are spurious. This is the primary criterion for a screening exercise in which some false positives are tolerable because survivors face further testing. Secondly, the Bonferroni Correction which controls the Family Wise Error Rate (FWER), the probability of any false rejection at all and is reported as a conservative benchmark.

Table 4.8: Pairwise Granger causality tests across all 20 ordered pairs (bivariate VAR(2)).

Pair	F	p	p (BH)	Sig
SPY $\rightarrow$ XOM	17.018	0.0000	0.0000	***
CVX $\rightarrow$ XOM	10.618	0.0000	0.0002	***
CVX $\rightarrow$ SPY	7.556	0.0005	0.0035	***
CVX $\rightarrow$ XLE	6.919	0.0010	0.0041	***
SPY $\rightarrow$ CVX	6.883	0.0010	0.0041	***
COP $\rightarrow$ XOM	6.186	0.0021	0.0069	***
COP $\rightarrow$ SPY	5.669	0.0035	0.0099	***
SPY $\rightarrow$ COP	5.490	0.0041	0.0103	**
CVX $\rightarrow$ COP	5.298	0.0050	0.0111	**
XLE $\rightarrow$ XOM	4.631	0.0098	0.0195	**
XLE $\rightarrow$ SPY	4.036	0.0177	0.0322	**
XOM $\rightarrow$ XLE	3.895	0.0204	0.0339	**
XOM $\rightarrow$ CVX	3.704	0.0246	0.0379	**
XLE $\rightarrow$ CVX	2.961	0.0518	0.0740	*
XOM $\rightarrow$ SPY	2.580	0.0758	0.1011	
SPY $\rightarrow$ XLE	2.198	0.1110	0.1388	
XOM $\rightarrow$ COP	0.975	0.3772	0.4437	
COP $\rightarrow$ CVX	0.474	0.6225	0.6917	
COP $\rightarrow$ XLE	0.203	0.8160	0.8589	
XLE $\rightarrow$ COP	0.014	0.9861	0.9861	

4.8.9.3. Results At the 10% significance level, 15 of the 20 pairs reject the null without adjustment. Under FDR, 14 pairs survive while only 8 survive the strict Bonferroni threshold.

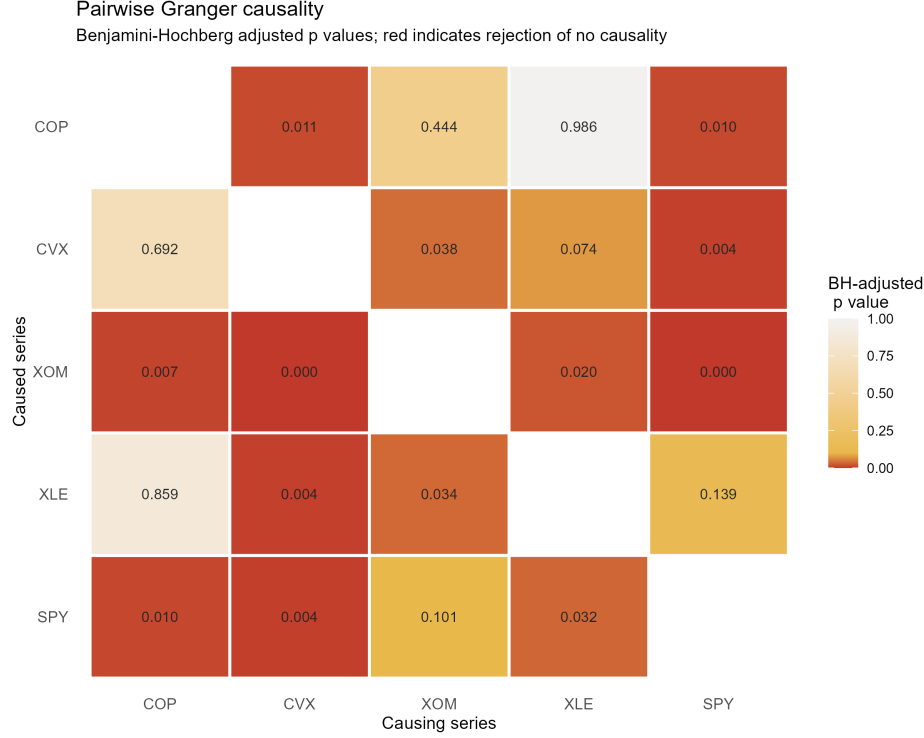


Figure 4.6: Benjamini and Hochberg adjusted p values for all twenty ordered pairs. Rows index the caused series and columns the causing series.

Ordered pair	Original, 2000 to 2018	This study, 2000 to 2026
XOM $\rightarrow$ COP	0.00010	0.377
COP $\rightarrow$ XOM	not reported as significant	0.00206

Table 4.9: Granger causality p values for the CP and XOM pair in the original sample and the extended sample. The direction of predictive content has reversed.

The original study selected XOM $\rightarrow$ COP as its primary backtesting pair driven by strong evidence ($p = 0.00010$). In the extended sample, this relationship completely collapses ($F = 0.975, p = 0.3772$). Additionally, the direction of predictive information flips to COP $\rightarrow$ XOM ($F = 6.186, p = 0.0021$)

One explanation for this may be due to business model changes of a firm. COP became a pure-play upstream (E&P) firm following its spin-off of Phillips 66 in 2012, whereas XOM

and CVX remained integrated majors. Upstream revenues depend directly on spot crude prices, whereas integrated firms hold downstream refining margins that expand when crude prices fall. This divergence in commodity risk exposure may have severed the historical lead lag linkage.

However, CVX underwent a reverse transformation. While CVX appeared exclusively as a caused series in the original sample, it acts as a primary causing series across the extended sample, significantly Granger-causing all four other instruments as seen in Figure 4.6. XOM and CVX now form the strongest bidirectional predictive loop among individual equities ($CVX \rightarrow XOM, p = 0.0002$; $XOM \rightarrow CVX, p = 0.0379$).

Table 4.10: Ordered pairs surviving Granger causality screening after Benjamini-Hochberg correction at the 10 percent level.

Pair	F	p	p (BH)
SPY $\rightarrow$ XOM	17.018	0.0000	0.0000
CVX $\rightarrow$ XOM	10.618	0.0000	0.0002
CVX $\rightarrow$ SPY	7.556	0.0005	0.0035
CVX $\rightarrow$ XLE	6.919	0.0010	0.0041
SPY $\rightarrow$ CVX	6.883	0.0010	0.0041
COP $\rightarrow$ XOM	6.186	0.0021	0.0069
COP $\rightarrow$ SPY	5.669	0.0035	0.0099
SPY $\rightarrow$ COP	5.490	0.0041	0.0103
CVX $\rightarrow$ COP	5.298	0.0050	0.0111
XLE $\rightarrow$ XOM	4.631	0.0098	0.0195
XLE $\rightarrow$ SPY	4.036	0.0177	0.0322
XOM $\rightarrow$ XLE	3.895	0.0204	0.0339
XOM $\rightarrow$ CVX	3.704	0.0246	0.0379
XLE $\rightarrow$ CVX	2.961	0.0518	0.0740

4.8.9.4. Final Candidate Pair Selection 14 ordered directional pairs survive the false discovery rate correction and are carried into the cointegration analysis of Section 5. Two qualifications govern how that set is used.

COP and XOM is retained despite the deterioration documented above. The pair is the one selected by the original study for its backtest and the central purpose of this replication is

to determine whether the relationship it identified survives the extended sample. Excluding the pair on the basis of a screening statistic would foreclose the question the paper exists to answer.

5. COINTEGRATION

5.1. Overview

While VAR in Chapter 4 was estimated on return, cointegration will instead use price levels directly. Two price series, each individually non-stationary are cointegrated when a linear combination of them is stationary, so that the gap between them is tethered to a fixed long run value and reverts to it after being disturbed. That reverting gap is the spread a pairs strategy trades and its existence is the precondition for everything in Chapter 6.

The Granger causality tests of Section 4.8.9 identify securities whose returns carry short horizon predictive information about one another. The IRFs showed that a shock to any of these series displaces its price permanently, so the levels wander without bound and no amount of return predictability guarantees that two prices stay close. Cointegration is the separate and stronger condition that some combination of two such wandering series does not wander. It is tested directly here.

5.2. The Engle and Granger Test

The two step procedure of Engle and Granger tests for cointegration through the stationarity of a regression residual. In the first step the log price of one security y is regressed on the log price of another security x ,

$$p_{y,t} = \tau + \beta p_{x,t} + e_t, \tag{5.1}$$

and the residual $\hat{e}_t$ is tested for a unit root. If the residual is stationary the two series are cointegrated, with β as the cointegrating coefficient (Hedge Ratio) and $\hat{e}_t$ as the spread. In the second step the dynamics of adjustment are estimated through the Error Correction

Mechanism (ECM):

$$\Delta p_{y,t} = \alpha + \gamma \hat{e}_{t-1} + \sum_{i=1}^k \phi_i \Delta p_{y,t-i} + \sum_{j=0}^m \delta_j \Delta p_{x,t-j} + \varepsilon_t, \quad (5.2)$$

where γ measures how rapidly disequilibrium is corrected (Adjustment Velocity). A statistically significant negative γ confirms an error correction mechanism and its magnitude indicates how fast disequilibrium is undone.

To ensure statistical correctness, two constraints must be addressed here. Firstly, Standard DF critical values (e.g., -2.86 at $\alpha = 0.05$ with an intercept) assume testing on an observed time series. However, applying ADF tests to an estimated residual $\hat{e}_t$ introduces explicit OLS optimisation bias. Since OLS minimises the sum of squared residuals ($\sum \hat{e}_t^2$), it inherently forces $\hat{e}_t^2$ to appear stationary as possible. This parameter optimisation artificially shifts the asymptotic distribution of the ADF test statistic to the left. Therefore evaluating OLS residuals against standard DF critical values results in severe Type I errors (over rejection of the null hypothesis of no cointegration). To correct for this parameter estimation effect, we utilize Engle-Granger (EG) critical values derived via Monte Carlo simulations. For a bivariate system with a constant term, the 5% critical threshold shifts leftward to approximately -3.34 .

Secondly, the procedure is also asymmetric. Because linear OLS minimises vertical rather than orthogonal residuals, normalising on p_y, t ($p_y \sim p_x$) yields a different slope coefficient and residual process than normalising on p_x, t ($p_x \sim p_y$) in finite samples:

$$\hat{\beta}_{y|x} \neq \frac{1}{\hat{\beta}_{x|y}} \quad (5.3)$$

To prevent arbitrary specification selection, I evaluate and report residual stationarity across both regression directions ($y \sim x$ and $x \sim y$).

5.3. Full sample tests

Table 5.1 reports the first step tests over the full sample for the candidate pairs carried from the Granger screening of Section 4.8.9.

Table 5.1: Full-sample Engle–Granger cointegration tests on log prices (2000–2026).

Pair	$\hat{\beta}$	ADF stat	Cointegrated at 5%
COP on CVX	0.9872	-3.208	No
XOM on XLE	0.9568	-3.142	No
CVX on COP	0.9572	-2.994	No
COP on XOM	1.2840	-2.912	No
XOM on COP	0.7296	-2.570	No
COP on XLE	1.2520	-2.437	No
CVX on XLE	1.2152	-1.992	No
CVX on XOM	1.2564	-1.908	No
XOM on CVX	0.7363	-1.768	No

No pair rejects the null of no cointegration over the full sample at the 5% level. The residual ADF statistics range from -3.208 for COP on CVX, the closest any pair comes to the Engle and Granger critical value of -3.34 , to -1.768 for XOM on CVX. COP on XOM, the pair the original study selected, returns -2.912 in the direction the original estimated and -2.570 in the reverse, both comfortably short of rejection.

This table could conclude that on 26 years of data, no candidate relationship is cointegrated and there is no spread to trade. Even so, I want to find out whether the original study’s own result can be reproduced on the extended sample.

5.4. The original result and the critical value

The original study reports COP and XOM as cointegrated over its estimation window of 31 December 1999 to 30 April 2013. Re-estimating the first step regression on exactly that window reproduces the residual ADF statistic: -3.159 for COP on XOM and -2.974 for the reverse.

Neither rejects the null of no cointegration at the 5% level against the Engle and Granger critical value of -3.34 . At -3.159 , however, the COP on XOM statistic does reject against the DF value of -2.86 and rejects comfortably against the no constant DF value of -1.94 . The original study's finding of cointegration is consistent with the application of a DF critical value to a residual that requires an Engle and Granger value. On the correct critical value the pair was not cointegrated even on the sample from which the original strategy was constructed.

This is the central result of the chapter. The relationship on which the entire original backtest was premised does not survive a correctly specified test of its own estimation sample. The underperformance of the strategy in the original's out of sample period, which the original reports but does not fully explain, follows naturally from a spread that was never stationary to begin with. A strategy that trades the reversion of a spread to its mean cannot succeed if the spread does not revert.

5.5. Rolling Cointegration

Evaluating cointegration over static multi year horizons returns a single metric that can mask time varying dynamics, intermittent mean reversion and structural regime shifts. To evaluate whether the asset pairs exhibit dynamic cointegration the first step Engle Granger regression is re estimated across a rolling window of 504 trading days ($\approx 2\text{years}$) advanced in monthly steps. The residual ADF statistic is recorded at each position and dated by the window's final day, producing the history of the statistic through time rather than a single

figure.

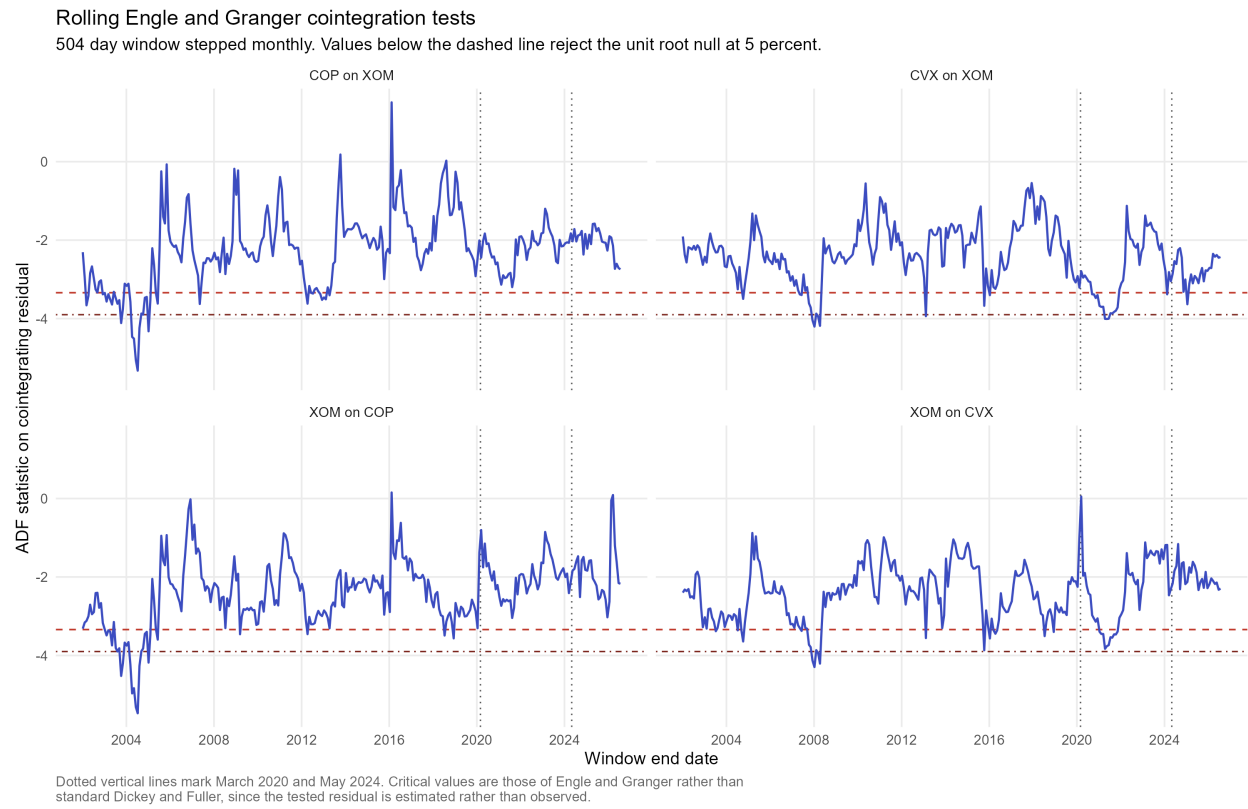


Figure 5.1: Rolling first step ADF statistics for four ordered pairs. A window lies below the dashed line when the pair tests as cointegrated at five percent against the Engle and Granger critical value. Dotted vertical lines mark March 2020 and May 2024.

Table 5.2:

Pair	Windows	Reject at 5%	Reject at 1%	Last rejection	Mean $\hat{\beta}$	SD $\hat{\beta}$
COP on CVX	295	12.9	4.4	2022-02-10	1.1372	0.5015
COP on XOM	295	11.9	2.4	2013-04-11	1.0790	0.6047
XOM on XLE	295	10.8	7.5	2018-07-12	0.7107	0.2896
XOM on CVX	295	10.5	1.4	2021-11-10	0.7701	0.3440
CVX on XOM	295	10.2	2.7	2025-01-16	0.8892	0.3629
CVX on XLE	295	9.8	3.7	2025-01-16	0.7658	0.2795
XOM on COP	295	9.8	3.1	2018-12-11	0.5513	0.3391
COP on XLE	295	7.5	1.7	2021-08-12	1.0516	0.3385

Across all eight pairs the share of windows rejecting the unit root null ranges from 7.5% to 12.9%. A persistently cointegrated pair would reject in the large majority of its windows. These marginal rejection rates show that mean reversion is occasional and transient rather than a stable structure of the relationship between two price series.

Importantly, for COP on XOM the most recent rejecting window ended on 11 April 2013. For more than a decade following this date, the estimated residual spread failed to reject the null hypothesis of non-stationarity in a single window, proving that the mean-reverting property vanished entirely post-2013.

5.6. Structural Reorganisation Around 2020

Splitting the rolling windows at March 2020 reveals that the pairs did not decay uniformly. They reorganised. Table 5.3 reports the rejection rate in each regime.

Table 5.3: Share of rolling windows rejecting the unit root null before and after March 2020. Engle and Granger 5 percent critical value: -3.34 .

Pair	Reject pre %	Reject post %	Change	Mean stat pre	Mean stat post
COP on XOM	16.1	0.0	-16.1	-2.253	-2.173
XOM on XLE	14.7	0.0	-14.7	-2.384	-2.152
XOM on COP	13.3	0.0	-13.3	-2.520	-1.959
COP on XLE	6.9	9.1	2.2	-2.173	-2.202
XOM on CVX	9.2	14.3	5.1	-2.440	-2.216
COP on CVX	11.5	16.9	5.4	-2.293	-2.266
CVX on XOM	6.0	22.1	16.1	-2.266	-2.764
CVX on XLE	5.5	22.1	16.6	-2.019	-2.793

The pattern is stark and it is not uniform across the securities. Three pairs fall to exactly zero rejections after March 2020. COP on XOM, from 16.1% to none; XOM on XLE, from 14.7% to none and XOM on COP, from 13.3% to none. Every one of these three involves XOM paired with COP or XLE. Over the same split, CVX and XOM orderings and CVX and XLE pairing move in the opposite direction with CVX on XLE rising from 5.5% to 22.1% and CVX on XOM from 6.0% to 22.1%.

The relationships involving COP did not merely weaken into noise. To the extent any relationship was present, it was replaced. Such marginal cointegration as existed among the integrated majors in the first half of the sample gave way to a somewhat stronger CVX and XOM association in the second while COP detached from the group entirely and never rejects the null again after early 2013.

This is the same conclusion reached by the Granger causality analysis of Section 4.8.9, arrived at through an independent route. That section found COP's short horizon predictive links to the sector collapsing and CVX inverting from a purely passive series into an active predictor of the others. The present section finds the same separation in the long run

relationship among price levels. Two tests with different null hypotheses, one on returns and one on prices agree on the structure of the change..

The economic reading is reinforced here. COP became a pure exploration and production company with the 2012 spin off of Phillips 66, while CVX and XLE remained integrated. The 2024 acquisitions deepened the divergence, with XOM absorbing Pioneer Natural Resources and COP absorbing Marathon Oil, each reshaping the constituent it enlarged. Securities that were once close substitutes evolved into different businesses.

5.7. Cointegrating Coefficient Instability

In addition to the non stationarity of the error term, the long run cointegrating parameter $\hat{\beta}$ exhibits severe time series instability.

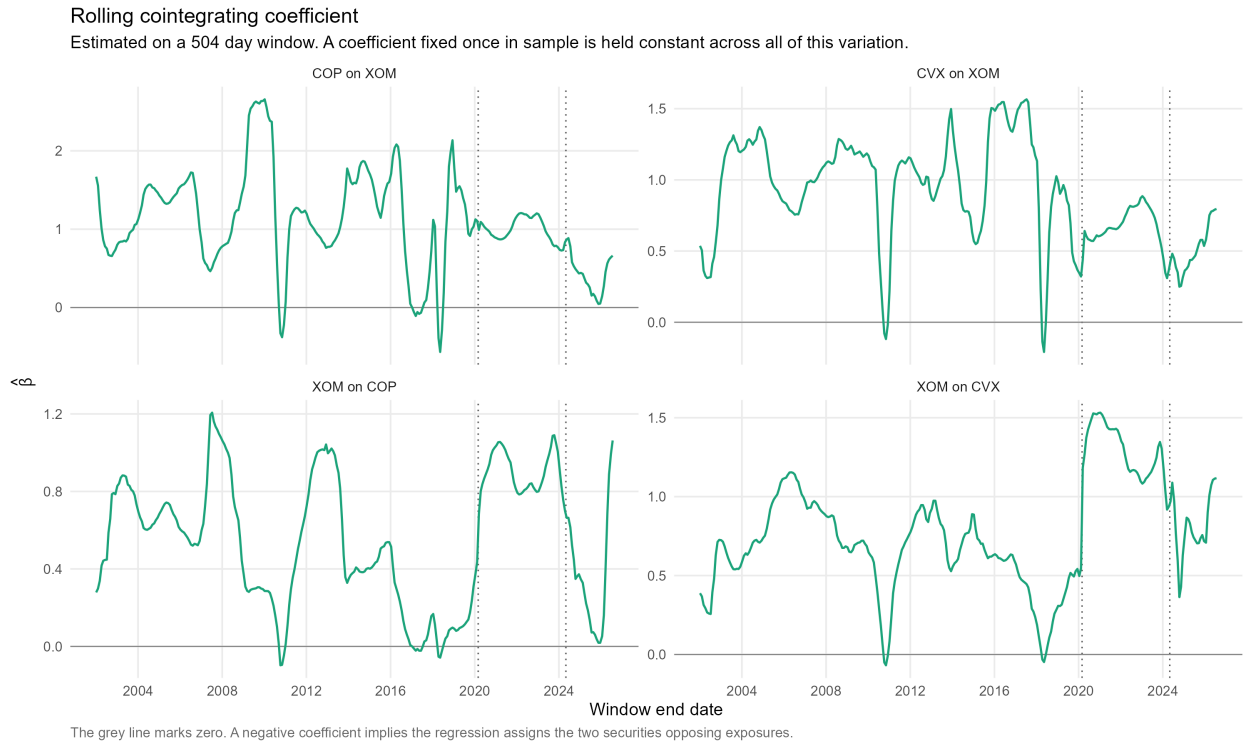


Figure 5.2: Rolling estimate of the cointegrating coefficient $\hat{\beta}$ for four ordered pairs. The grey line marks zero.

For COP on XOM, the coefficient ranges across the sample from above 2.5 to below 0.0, with a standard deviation of 0.60 around a mean of 1.08, as recorded in Table 5.2. The negative coefficients are the more interesting to observe. A negative cointegrating coefficient ($\hat{\beta} < 0$) implies that the portfolio hedge assigns opposing directional positions to asset legs that move together, forcing a pairs-trading model to take simultaneously long or simultaneously short positions on both securities rather than maintaining a market-neutral spread.

5.8. Selection for Backtesting

The evidence of this chapter does not support any pair as cointegrated, either over the full sample or on the original study's own estimation window when the correct critical values are applied. For the purpose of replication, the strategy of Chapter 6 is nonetheless applied to COP and XOM, the pair selected by the original.

This choice is made deliberately and in full knowledge of the preceding results. The object of the exercise is to determine how a strategy premised on a cointegrating relationship performs when that relationship is absent as observed in this section. The expected outcome is that a spread strategy built on a spread which does not reliably mean revert will not reliably profit, is itself the substantive finding of the replication.

6. STRATEGY BACKTESTING

6.1. The strategy

The trading strategy evaluates mean reversion signals on the cointegrating residual s_t derived from the log prices of COP ($p_{COP,t}$) and XOM ($p_{XOM,t}$):

$$s_t = p_{COP,t} - \tau - \beta p_{XOM,t}, \quad (6.1)$$

where the intercept $\tau = -1.3245$ and hedge ratio $\beta = 1.2561$ are held constant at their insample estimates mirroring the original study's methodology. Since the estimated cointegrating coefficient β is positive, the spread s_t defines a long/short market neutral basket. To make entry and exit thresholds scale invariant, the spread is standardised into a z -score (z_t) using the Ornstein-Uhlenbeck in sample equilibrium mean (μ_e) and stationary standard deviation (σ_{eq}):

$$z_t = \frac{s_t - \mu_e}{\sigma_{eq}}. \quad (6.2)$$

Trading decisions are driven by standard thresholds crossings:

- Short Spread Position: When $z_t > +\Delta$ (indicating COP is overvalued relative to XOM), a short spread position is initiated by shorting COP and longing XOM
- Long Spread Position: When $z_t < -\Delta$ (indicating COP is undervalued relative to XOM), a long spread position is initiated by longing COP and shorting XOM
- Exit Position: Open positions are closed when the z -score crosses back through zero ($z_t = 0$), realising profits as the spread reverts to its equilibrium mean μ_e

To account for market microstructure noise and execution delays, trade execution is lagged by k periods following a signal trigger, serving as a proxy for slippage. The Entry threshold Δ and the execution lag k , the decision parameters, are selected via a grid search optimising

the in sample annualised sharpe ratio. The optimal specification (Δ^*, k^*) is then applied unchanged across both out of sample and extension evaluation periods.

6.2. The Ornstein-Uhlenbeck Parameters and Implications

Before any trade is simulated, the estimated OU parameters reveal a flaw in the strategy's design. Table 6.1, reports the in-sample parameter estimates derived from the COP–XOM spread.

Table 6.1: In-sample Ornstein–Uhlenbeck spread parameters

Parameter	Value
Cointegrating coefficient $\hat{\beta}$	1.2561
AR(1) coefficient B	0.99504
Reversion speed θ (per day)	0.00498
Equilibrium level μ_e	0.01478
Stationary sd σ_{eq}	0.13517
Half-life (days)	139.3
Half-life (trading years)	0.55

The estimated speed of mean reversion is $\theta = 0.00498$ per day and AR(1) coefficient ($B = 0.99504$). This implied a half life of 139.3 trading days approximately 0.55 trading years or nearly seven calendar months.

The half life represents the time required for the process to decay by 50% back toward its long term mean μ_e . A daily threshold based mean reversion strategy operates on an implicit premise that states that the deviations z_t are transient shocks that decay rapidly, allowing a trader to enter, capture the convergence and free up capital over a short horizon of days or weeks. A 139 day half life completely undermines this mechanism.

The strategy's calibration thus contains a fatal contradiction which is the model used to

justify trading outputs a half life that invalidates the holding period logix required to execute the trade. Instead of capturing rapid and transient mispricings, any strategy built on this spread is forced into a long horizon trade exposing capital to extended periods of drift and capital lockup (opportunity costs).

6.3. Replication results

Parameter optimization via in-sample grid search yields an optimal entry threshold of $\Delta^* = 2.5\sigma_{eq}$ and an execution lag of $k^* = 2$ days (In-sample Sharpe: 0.70). Table 6.2 reports the performance of this specification across all four periods against two benchmarks: a buy-and-hold of the spread and the XLE Energy ETF.

Table 6.2: Performance of the fixed-parameter COP–XOM strategy across periods.

Period	Metric	CAGR	Vol	Sharpe	Sortino	MaxDD
In-sample	Strategy	8.05%	12.06%	0.703	1.067	21.14%
In-sample	Buy & hold	-0.44%	21.38%	0.087	0.120	60.81%
In-sample	Benchmark (XLE)	5.26%	29.75%	0.322	0.443	64.83%
Out-of-sample	Strategy	8.35%	14.74%	0.617	1.014	23.06%
Out-of-sample	Buy & hold	2.97%	22.00%	0.243	0.346	54.54%
Out-of-sample	Benchmark (XLE)	-4.77%	20.40%	-0.137	-0.189	49.25%
Extension (2019-2026)	Strategy	3.46%	17.41%	0.283	0.408	34.34%
Extension (2019-2026)	Buy & hold	-9.53%	22.79%	-0.325	-0.463	61.39%
Extension (2019-2026)	Benchmark (XLE)	9.12%	32.57%	0.433	0.593	66.35%
Total	Strategy	6.79%	14.34%	0.530	0.799	34.34%
Total	Buy & hold	-2.41%	21.92%	-0.002	-0.002	68.21%
Total	Benchmark (XLE)	4.09%	28.89%	0.285	0.391	79.27%

Surprisingly, the strategy beats buy-and-hold of the spread in every period on a risk-

adjusted basis and beats the XLE benchmark in three of the four. Over the full sample it returns a Sharpe ratio of 0.530 against the benchmark's 0.285 and in the out-of-sample period it returns 8.35% annually against the benchmark's loss of -4.8% annually.

The trading strategy generates positive performance despite the complete absence of full sample cointegration is a highly surprising result. To reconcile these findings, I did more research on the formulas to understand where the edge comes from.

6.3.1. The Edge is Variance Reduction

The strategy's advantage over the benchmark is concentrated in the denominator of the Sharpe ratio rather than the numerator. Its annualised volatility ranges from 12.06% to 17.41% across periods, against the benchmark's 20.40% to 32.57%. A spread position is long one security and short a multiple of another, so the common component of their returns cancels and the position is mechanically low variance. This variance reduction is a property of holding a hedged position and is independent of whether the spread reverts. The strategy is rewarded for market neutrality, which it genuinely provides, rather than for mean-reversion, which the spread does not reliably offer.

The clearest evidence of this is in the drawdowns. Over the full sample the strategy's maximum drawdown is 34.34% against 68.21% for the unhedged spread and 79.27% for the sector fund. The strategy loses roughly half as much at its worst as the benchmark it is measured against. This is the signature of a hedge: not a higher return but a materially shallower loss when the market moves against the position. The Sharpe advantage documented above is this same property expressed as a ratio, a moderate return divided by a low volatility, rather than evidence of a return generating edge.

6.4. Interpretation

The weakening edge is the expected consequence of a relationship that was cointegrated intermittently and weakly, in the in sample and has not rejected the null of no cointegration in

a single rolling window since April 2013. A strategy premised on daily statistical arbitrage should not be expected to perform once the very property it depends on, established in Section 5.5 to have vanished.

Across a regime containing the 2020 collapse in crude prices, 2022 Russia-Ukraine war and the 2026 Iran war, the strategy maintains a positive annualised return of 3.46% and restricted its maximum drawdown to 34.34%, against a 68.21% drawdown in the unhedged spread and 79.27% in the sector fund.

These two readings are reconciled by recognising what the strategy actually is. It does not function as the continuous mean-reversion engine its design implies. The Ornstein-Uhlenbeck half-life of 139 days, established in Table 6.1, already indicated that the spread does not revert on the horizon the threshold rules assumptions and the cointegration tests of Chapter 5 indicated that it may not revert reliably at all. What the strategy does instead is hold a market neutral relative value position whose two legs offset most of the common energy-sector movement. That offsetting is mechanical. It delivers variance reduction and downside protection whether or not the spread satisfies the formal econometric conditions for statistical arbitrage.

Therefore, the strategy does not generate excess return, its risk-adjusted performance rests on a denominator effect rather than a numerator effect and in a rising sector it will underperform a simple long position, as the extension period shows. However, a characterisation of the strategy as merely broken, which the cointegration results in isolation might suggest, understates what a market-neutral construction provides even in the absence of the reversion it was designed to exploit.

Whether the risk-adjusted return that remains constitutes genuine skill or merely compensation for exposure to common risk factors such as low volatility and value, cannot be settled by the performance statistics alone. That question requires decomposing the strategy's returns against a factor model, which is the subject of Section 7.

7. RISK FACTOR ATTRIBUTION

7.1. Question

Chapter 6 showed that the strategy earns a positive risk-adjusted return whose most visible source is the variance reduction inherent in a hedged position. However, raw performance metrics and Sharpe ratios leave a critical quantitative question unanswered: after stripping away this volatility dampening, does the strategy generate true residual excess return (α) that common asset-pricing risk factors cannot span? Furthermore, if such an edge exists, is it a durable feature of the spread relationship or merely a statistical edge driven by strong performance in the earlier portion of the sample?

To isolate true alpha from passive risk factor compensation, this chapter decomposes the strategy's daily excess returns using the Fama and French five-factor model augmented with Carhart's momentum factor. The model is estimated first over the full sample and then within each sub period (in sample, out of sample and extension.

7.2. Model Specification

Daily strategy excess returns ($r_t - r_{f,t}$) are regressed on the market, size, value, profitability and investment factors of Fama and French [13], together with Carhart's momentum factor [14].

$$r_t - r_{f,t} = \alpha + \beta_{\text{MKT}} \text{MKT}_t + \beta_{\text{SMB}} \text{SMB}_t + \beta_{\text{HML}} \text{HML}_t + \beta_{\text{RMW}} \text{RMW}_t + \beta_{\text{CMA}} \text{CMA}_t + \beta_{\text{MOM}} \text{MOM}_t + \varepsilon_t. \quad (7.1)$$

The intercept α measures the average daily return unexplainable by linear combinations of the six systemic risk factors. A statistically significant $\alpha > 0$ indicates a true edge whereas $\alpha = 0$ implies the strategy's return are merely a repackaging of stand equity risk premium.

Since the strategy exhibits a long half life and holds positions across multi month windows,

daily portfolio holdings are highly overlapping. This will induce serial autocorrelation in the residual term ϵ_t . Standard OLS standard errors severely underestimate true parameter variance under serial correlation, artificially inflating t -statistics and overstating statistical significance.

To resolve this standard errors are estimated using the Newey and West covariance matrix with 10 lags. This bandwidth choice aligns with standard empirical rule-of-thumb selection criteria ($m \approx \lfloor 4 \times (T/100)^{2/9} \rfloor$), capturing two calendar weeks of trading memory. This choice accounts for heteroskedasticity and adjusts standard errors across 10 trading days of autocorrelation.

7.3. Full sample attribution

Table 7.1: Factor attribution of daily strategy excess returns across factor specifications.

Factor	CAPM	FF5	FF5 + MOM
Alpha (daily)	0.0002**	0.0002**	0.0003**
Market	0.0086	-0.0041	-0.0167
SMB (size)		-0.0236	-0.0234
HML (value)		0.0855**	0.0333
RMW (profitability)		-0.0889**	-0.0855**
CMA (investment)		0.0037	0.0528
MOM (momentum)			-0.0883***
R^2	0.000	0.006	0.015

Across all three models, the intercept (α) is positive and significant. Contrary to the standard expectation that controlling for risk premium would absorb portfolio returns, the annualised factor adjusted alpha expands as controls are added, rising from 5.04% under CAPM to 6.34% under the full six factor model.

Across all specifications, the market loading is economically negligible and statistically indistinguishable from zero ($\beta_{MKT} = -0.0167, p = 0.39$).

The momentum loading represents the most statistically dominant coefficient in the model ($\beta_{MOM} = -0.0883, p = 2.3 \times 10^{-6}$). A strongly negative momentum loading is the definitive factor signature of a mean-reverting quantitative strategy. By construction, the algorithm opens long positions in the underperforming security (the relative "loser") and short positions in the outperforming security (the relative "winner").

The profitability loading is negative and statistically significant ($\beta_{RMW} = -0.0855, p < 0.05$). In an energy pair consisting of an integrated major and a pure-play exploration and production (EP) firm, this tilt reflects structural corporate differences: the algorithm frequently holds short positions in the higher-margin, integrated major (ExxonMobil) while going long the higher-beta pure-play EP (ConocoPhillips) during spread expansions.

The value loading (HML) is positive and significant under the five-factor model ($\beta_{HML} = 0.0855, p < 0.05$), but loses significance once momentum is added ($\beta_{HML} = 0.0333$). This shift is a direct outcome of the negative cross-sectional correlation between value and momentum factors, with momentum absorbing the shared covariance.

However, the statistically significant 6.34% annualised alpha appears to challenge the central thesis of preceding chapters stating that the *COP-XOM* price relationship lacks a persistent cointegrating bond. We then try disaggregating the time periods in order to examine the alpha generation.

7.4. Factor Attribution by Period

The full-sample alpha is an average over the whole sample, and Chapter 6 has already shown that the strategy's returns were not uniform across it. Estimating Equation (7.1) separately within each period reveals whether the alpha shares that concentration.

Table 7.2: Factor-adjusted alpha and momentum loading by period

Period	Obs.	Alpha (ann.)	95% CI	MOM loading	R^2
In-sample	3352	7.36%**	[1.6, 13.1]	-0.053**	0.020
Out-of-sample	1428	10.33%*	[-1.2, 21.9]	-0.207**	0.109
Extension	1883	3.17%	[-8.1, 14.4]	-0.102***	0.017
Full sample	6663	6.34%**	[1.3, 11.4]	-0.088***	0.015

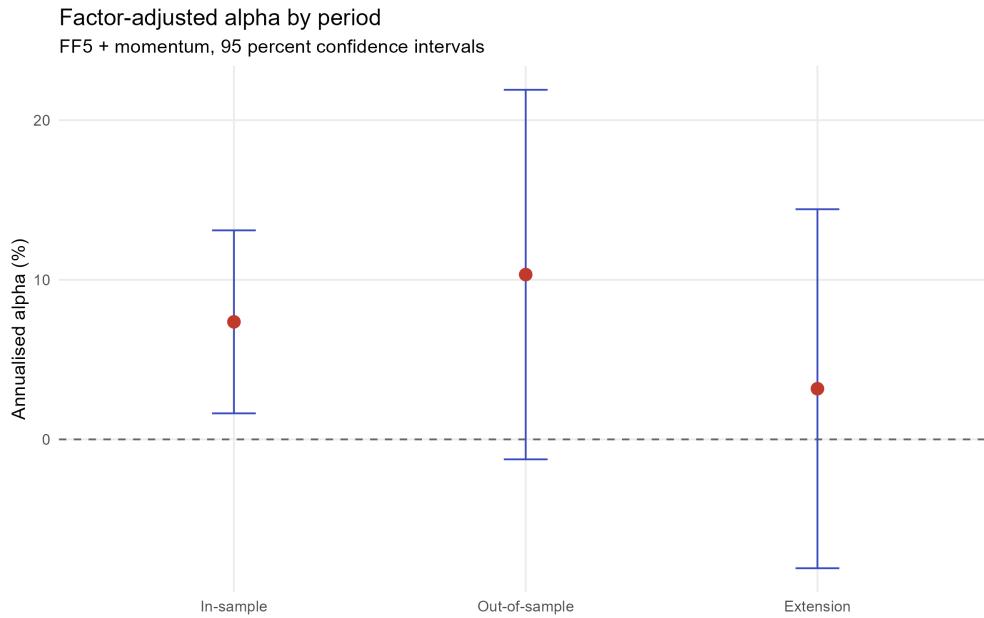


Figure 7.1: Annualised factor-adjusted alpha within each period, with 95 percent confidence intervals. The in-sample alpha is significantly positive; the extension alpha is statistically indistinguishable from zero.

In the in-sample period, the strategy generated an annualised factor adjusted alpha of 7.36% ($p = 0.012$), with a 95% confidence interval bounded strictly above zero ([1.60%, 13.12%]). During this regime, the market actively rewarded the pair's mean-reversion signals, validating the original study's claim of an exploitable statistical arbitrage opportunity.

In the initial out of sample period, the point estimate for annualised alpha reached its peak at 10.33%. However, this estimate carries a elevated standard error and becomes only

marginally significant ($p = 0.081$), reflected in a wide confidence interval spanning negative territory ($[-1.24\%, 21.90\%]$). This widening uncertainty captures the growing lumpiness and path dependency of strategy returns as spread holding periods began to stretch.

In the extension period, the annualized factor-adjusted alpha collapses to 3.17% with a p -value of 0.580. The 95% confidence interval ($[-8.06\%, +14.40\%]$) easily spans zero. Across 1,883 daily observations—a sample size large enough to reject small-sample sample noise—the strategy’s alpha becomes statistically indistinguishable from zero.

7.5. Implications

Taken together, the sub period disaggregations reveal a disappearing alpha and stable momentum. The negative momentum exposure confirms that the strategy’s executions were uniform through out the multi-decade backtest. The algorithm continuously entered contraction and mean reverting positions. Since the execution rules were held constant, the decay in excess returns cannot be attributed to algorithmic errors, data artifacting or shifting execution parameters. Instead, the loss of alpha directly mirrors the structural breakdown in spread cointegration documented in Chapter 5. As market arbitrage efficiency tightened, historical price divergence ceased to guarantee subsequent mean reversion, converting a once-profitable statistical arbitrage strategy into a uncompensated risk factor.

8. CONCLUSION

8.1. Outcome of Research Objectives

This paper replicated and in a different programming language, a cointegration based pairs trading study of energy equities and extended its sample forward by roughly seven and a half years. The pedagogical aim was to engage with every methodological choice a reader would otherwise skim. The substantive aim was to test whether the relationship the original identified and the strategy it built on that relationship survive on data the original author could not have seen. The extended period was not chosen for convenience. It spans the 2020 collapse in crude oil prices, the 2024 consolidation of United States shale assets and the 2026 Iran war and it therefore subjects the original's conclusions to precisely the kind of structural change under which a claimed long-run relationship is most likely to fail.

8.2. What the strategy is, in the end

The strategy is not the continuous mean-reversion engine its design implies and on the evidence of this paper it never was. What it is instead is a market-neutral relative-value allocation whose two offsetting legs deliver genuine downside protection irrespective of whether the spread reverts. In a period of severe energy market dislocation it preserved capital far better than either an unhedged spread or a long sector position, and that is a real and useful property. But it is a different property from the one the strategy was built to exploit and the statistical arbitrage the construction was designed to capture is, in these securities and over this sample, no longer present.

8.3. Limitations and extensions

Several limitations bound these conclusions and mark the natural directions for further work. The backtest is gross of transaction costs which is a contrarian strategy trading on threshold crossings turns over frequently and the modest alpha of the earlier sample would

be thinner net of the bid-ask and financing costs the model omits. The analysis holds the cointegrating coefficient fixed, as the original does. Finally, the securities studied are a small and deliberately homogeneous set; the reorganisation this paper documents among them invites a broader examination of how the integrated and pure-play segments of the energy sector have diverged over the period, of which the ConocoPhillips separation is likely one instance among many.

APPENDIX SECTION

A.1. Full lag order selection criteria

Table A.1 reports the value of each information criterion for every lag order from 1 to 20, underlying the selection discussion in Section 4.7.

Table A.1: Information criteria by lag order, $p = 1$ to 20				
Lag	AIC	HQ	SC	FPE
1	-46.179083	-46.168494	-46.148431	0.0000000000
2	-46.183745	-46.164332	-46.127549	0.0000000000
3	-46.184343	-46.156107	-46.102605	0.0000000000
4	-46.190426	-46.153365	-46.083144	0.0000000000
5	-46.191495	-46.145610	-46.058669	0.0000000000
6	-46.191438	-46.136730	-46.033069	0.0000000000
7	-46.197123	-46.133590	-46.013211	0.0000000000
8	-46.196042	-46.123685	-45.986586	0.0000000000
9	-46.197694	-46.116513	-45.962695	0.0000000000
10	-46.197373	-46.107369	-45.936831	0.0000000000
11	-46.195806	-46.096978	-45.909720	0.0000000000
12	-46.193248	-46.085595	-45.881619	0.0000000000
13	-46.193829	-46.077353	-45.856657	0.0000000000
14	-46.189325	-46.064025	-45.826609	0.0000000000
15	-46.197227	-46.063102	-45.808968	0.0000000000
16	-46.194847	-46.051899	-45.781045	0.0000000000
17	-46.193316	-46.041544	-45.753971	0.0000000000
18	-46.192856	-46.032260	-45.727967	0.0000000000
19	-46.189285	-46.019865	-45.698853	0.0000000000
20	-46.187199	-46.008955	-45.671223	0.0000000000

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